Phase-Scoped Query Layouts for Reference-Preserving Speculative Decoding

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Abstract

Numerical repair changes how much batching speculative decoding can exploit. We show that repair scope can change whether drafting reduces request time. Phase-scoped query-to-batch (QB) attention uses paired query/mask views and shared K/V storage to retain one-query attention organization alongside batched projections and feed-forward computation. Phase dispatch and layout-sensitive graph reuse integrate this representation with the speculative transaction. On a matched seven-arm CUDA panel, local attention repair makes drafting profitable for Qwen3 and Llama, where global serialization makes it costly. Both explicit per-query attention (PQ) and QB cross this boundary. SmolLM2 benefits from reference-preserving target-only batching but remains faster without drafting. A retrospective E2 analysis on the same exposed panel retains all three local-policy directions across six sessions on two dates. QB separately reduces speculative resident-request latency over PQ by 3.06%, 6.17% and 2.83% across the three families; 72 captured contexts show bitwise PQ/QB agreement at QK, softmax and AV. A matched Qwen3 study finds 6.3% and 5.4% lower complete service latency at $k = 3$ with resident and fresh contexts; neither $k = 1$ cell clears the practical screen. New-input tests qualify Qwen3 and SmolLM2 but expose a Llama case where both repairs depart from the deployed reference. State interventions and a Vulkan port identify the corresponding producer-state and backend requirements. Together, these results make repair scope, draft choice and complete service cost explicit decisions within a qualified workload.

Keywords: speculative decoding, execution geometry, numerical consistency, runtime graph transformation, state composition

1. Introduction

Speculative decoding saves sequential target invocations by verifying several draft tokens together [1, 2]. That change also alters the target's execution geometry. Floating-point evaluation can depend on query width, so mathematically equivalent batching can change the greedy response of a deployed quantized model. When an application must retain its existing regression baseline, recovering the reference output becomes part of the optimization problem.

The decisive choice is where to impose that numerical constraint. Global serialization (GS) evaluates one target row at a time through projections, attention and feed-forward layers, giving up the surrounding batching. Per-query attention (PQ) confines the separation to attention but constructs a subgraph for each query. We ask how this choice of repair scope changes the economic case for drafting, and how to express local repair at lower cost.

Phase-scoped query-to-batch attention (QB) implements local repair through graph lowering. It views Q as $(d, 1, h, q)$ instead of $(d, q, h, 1)$ and transforms the mask in parallel. Existing attention operators process the queries as separate batch instances while K/V storage remains shared within one sequence. Projections and feed-forward computation stay batched. Role/phase guards and a layout-sensitive graph cache bind this representation to the native speculative transaction.

The contribution is this runtime contract and its measured consequences. Query independence itself is standard attention structure. Batch-invariant kernels control reductions [3]; CoRun controls prefill and decode shapes [4]; LLM-42 verifies a fast path under a fixed-shape schedule [5]. QB preserves a local one-query attention organization through views, head mapping, phase dispatch and graph reuse. Section 2 defines its workload qualification, and Section 9 compares the control points.

The matched CUDA experiments distinguish repair scope, draft choice and representation cost. Local repair changes drafting from a net cost to a net benefit for Qwen3 and Llama; both PQ and QB cross this boundary. A retrospective draft-on/off analysis of the separately scheduled E2 study retains these local-policy directions in all six sessions: drafting benefits Qwen3 and Llama, while SmolLM2 remains faster without it. Within local repair, QB reduces speculative resident-request latency over PQ by 3.06%, 6.17% and 2.83% across the three families. A 72-context replay finds bitwise agreement at QK, softmax and AV. Target-only QB also beats the serial reference on every task of the original matched panel, providing a useful choice when the draft adds cost.

The matched deployment study tests whether the layout benefit survives complete service costs. One Qwen3 runtime crosses $k = 1, 3$ with resident and fresh contexts. At $k = 3$, QB reduces complete service latency by 6.3% and 5.4%, respectively; neither $k = 1$ cell clears the practical screen [6]. The larger-draft benefit therefore survives context creation and teardown (Section 6.1).

Transfer experiments identify the interfaces that qualification must cover. Vulkan state interventions show

that endpoint-equivalent states can contain noninterchangeable KV components. A grouped-query port preserves the existing AV evaluation while dispatching independent queries. On previously unexposed natural inputs, Qwen3 and SmolLM2 retain the reference, while a Llama input makes PQ and QB diverge together from the deployed reference [7]. Their mutual agreement therefore cannot certify a new input's reference behavior (Section 7.1).

The contributions are:

(1) A phase-scoped attention transformation with paired Q/mask views, shared-KV and head semantics, ordered output recovery and layout-sensitive graph reuse.

(2) Matched evidence that repair scope changes draft profitability, with QB's incremental PQ-relative benefit measured separately in resident requests and complete service lifecycles.

(3) State, backend and new-input tests that establish concrete qualification requirements and expose failures of local-to-request composition.

The reproduction records include an RTX core/frontend rebuild with pinned numerical DLLs and a B570 numerical-library/shader rebuild, each checked by native stage and request replay on its source host [6, 8]. The supplement and ancillary files supply the complete comparisons and implementation identities [9].

2. Reference contract and execution model

2.1. Output qualification

Let ρ identify the reference model and tokenizer bytes, prompt construction, numerical runtime, ordinary execution recipe, sampler and stopping rule. For input x, its endpoint is

$$O_\rho(x) = (\mathbf{y}, \mathbf{e}, n, y_{\text{last}}, s), \tag{1}$$

where $\mathbf{y}$ is the full native token-ID array, $\mathbf{e}$ its end-of-generation (EOG) flags, n its length and s the stop reason. The principal campaigns stop at native EOG or 128 generated tokens. A candidate qualifies on a registered panel X when every required pair has $O_c(x) = O_\rho(x)$, the reference is stable and all required records are present. Latency comparisons use these qualified, complete responses.

We distinguish exact-arithmetic equivalence, bitwise operator correspondence and endpoint agreement. Each answers a different question. Backend acceptance and actual operator placement are additional execution prerequisites. These definitions apply throughout the paper; qualification always refers to the stated configuration, panel and output window.

2.2. Workload qualification and use

The study selects implementations for declared, replayable workloads under a fixed reference. Three domains have distinct roles. The *backend-admitted domain* comprises configurations accepted by the source and operator guards. The *validated domain* comprises the inputs and output windows that pass complete-reference comparison under those configurations. The *use domain* is the workload for which an implementation is selected after qualification. Admission alone does not certify numerical correspondence; the later Llama counterexample lies within the admitted configuration (Section 7.1).

For a workload requiring the deployed reference, the decision rule retains R for inputs outside the validated domain and compares qualified candidates within it. Extending that domain requires fresh reference checks. This is an offline selection rule: the evaluated implementation has phase and configuration guards, but no online detector or recovery guarantee for arbitrary new inputs.

2.3. The stateful transaction

A target context owns KV values, sequence positions, a pending sampled token or available logits, and relevant sampler state. The draft has separate state. It proposes up to k tokens; the target evaluates the pending token and proposal positions; acceptance, correction and any bonus determine the committed prefix. Rejected positions are reconciled before the next cycle. A visible token may still be pending evaluation into KV and must be consumed exactly once.

QB changes attention representation within the ordinary speculative verification and reconciliation transaction. Microbatch capacity U, occupied query count q, caller width and output reservation remain distinct. We write UBn for a configured capacity of n. A caller containing $k+1$ rows can execute as several physical micrographs, so caller counts alone do not describe parallel work.

2.4. Local repair and state composition

The design objective is to minimize complete request time over tested implementations, subject to endpoint qualification. Write a request fragment as $D_g(P_g(x))$, where P_g forms the state consumed by D_g under execution organization g. Correspondence of D_{QB} and the reference on one input state s_ρ does not establish correspondence on the different state $P_g(x)$.

For full stateful composition, a relation between candidate and reference states must hold initially, preserve the committed output at each transaction, and hold again after reconciliation. This is a standard sufficient-condition argument; its premises are tested empirically here. Section 7 examines two simpler assumptions often used in local reasoning: that endpoint-equivalent states have interchangeable components, and that enlarging a sufficient replacement preserves its effect.

3. Phase-scoped query-layout design

3.1. Separating attention from target-wide batching

Figure 1 contrasts the three repair organizations. GS processes target rows one at a time throughout projection, attention and the feed-forward network. PQ retains batched surrounding work but explicitly constructs an attention subgraph for each query and concatenates the results. QB retains that local one-column organization through a view: queries become separate batch instances of the existing attention operators. It does not serialize the surrounding layers or construct a distinct attention subgraph per query.

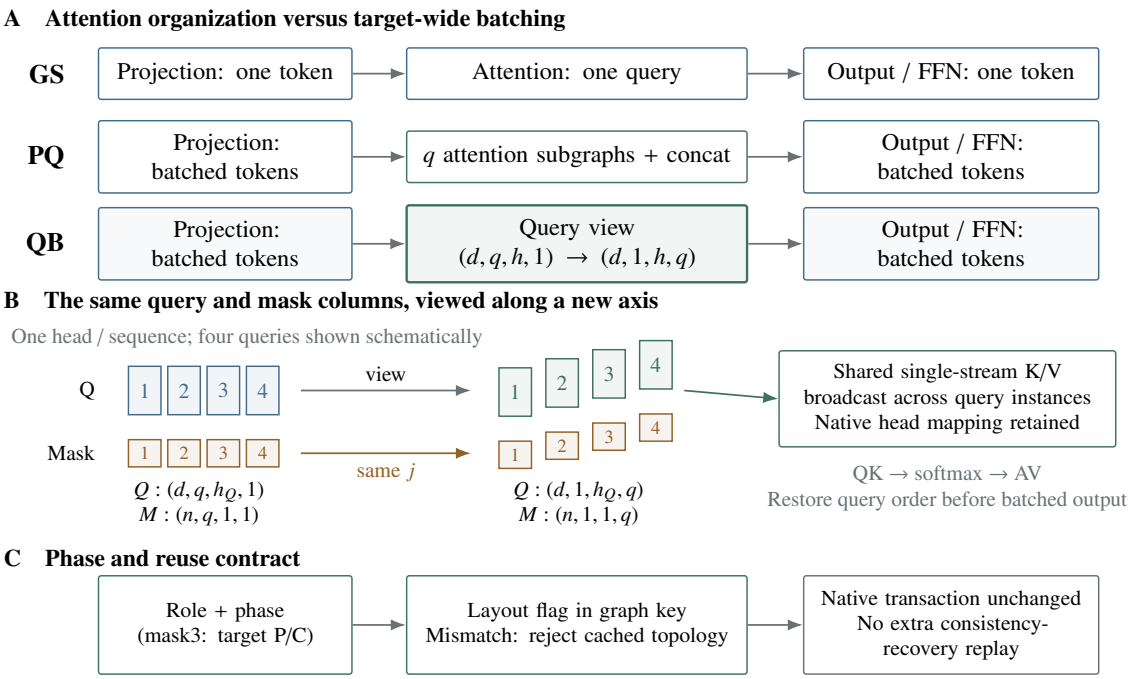


Figure 1: Three same-reference implementation alternatives. Boxes show logical graph organization, not measured kernel occupancy. QB changes attention representation while keeping surrounding target computation batched. Numbered query and mask tiles depict logical indexing, not copied K/V or independent sequences. Phase selection is part of graph construction and reuse; the speculative transaction is unchanged.

We label the ordinary graph with multiple query positions MQ; this denotes execution geometry, not the multi-query attention (MQA) architecture with one shared K/V head. The intervention point follows the native Q/K/V permutations in the non-flash attention builder. Let d be the query/key channel dimension, q the number of occupied queries, h_Q the query-head count, and n the key extent. Using the runtime's axis order, the input query has shape $(d, q, h_Q, 1)$. QB creates a four-dimensional view with shape $(d, 1, h_Q, q)$ and preserves the original query-column stride as the new fourth-axis stride. This is a view of the same input values, not a different projection or a re-encoding of the tokens.

3.2. Address, mask and head semantics

Write the original byte strides as s_0, s_1, s_2, s_3 and the base pointer as b_Q. For channel i, head a and logical query j, the original address is

$$\mathrm{addr}_{\mathrm{MQ}}(i, j, a, 0) = b_Q + is_0 + js_1 + as_2. \tag{2}$$

The transformed view satisfies

$$\mathrm{addr}_{\mathrm{QB}}(i, 0, a, j) = b_Q + is_0 + as_2 + js_1, \tag{3}$$

so it selects the same scalar. The singleton query-column stride is immaterial for access, but the implementation still supplies a valid native view descriptor. It must not invent a contiguous layout when the source is strided.

The mask is transformed in parallel from $(n, q, 1, 1)$ to $(n, 1, 1, q)$. For original mask strides m_0, m_1, query j still reads $b_M + tm_0 + jm_1$ for key index t. The original mask values, including padded or disallowed positions, are retained. Changing the view must not replace a query-dependent causal mask by a single shared column. Logical token positions are unchanged: the new batch coordinate labels queries in one sequence, not independently positioned sequences.

Let $\kappa(a)$ denote the native grouped-query head mapping from query head a to a K/V head. Under an integral grouped-query ratio, the same mapping is used by all three organizations. K and V retain their native values, strides and single-stream ownership and broadcast over the new query-batch coordinate. In mathematical notation, for the already implemented scaling α and admissible mask m,

$$z_{tja} = \alpha \sum_i K_{it\kappa(a)} Q_{ija} + m_{tj}, \tag{4}$$

$$p_{tja} = \frac{\exp(z_{tja})}{\sum_u \exp(z_{uja})}, \tag{5}$$

$$O_{vja} = \sum_t V_{vt\kappa(a)} p_{tja}. \tag{6}$$

These equations describe logical indices; the actual runtime may store or permute V differently. QB must retain that implementation's mapping rather than reinterpret V by this notation alone.

Proposition 1 (Mathematical view equivalence)**.** *For a supported single-stream attention graph, suppose Equation (3) and the corresponding mask mapping hold, K/V broadcasting preserves κ, operator parameters are unchanged, and output recombination restores query order. In exact arithmetic, QB and the original attention graph compute the same logical O in Equation (6).*

Proof. For each fixed (j, a), both representations select identical Q, K, V and mask values. The sums range over the same channel and key indices, and softmax is normalized over the same keys. The batch label in QB separates query instances but changes neither their operands nor their logical dependence. Recombination restores (v, j, a) order.

Thus every logical output value agrees in exact arithmetic. Finite-precision correspondence is established separately by the operator and complete-output experiments. □

After QK, softmax and AV, the native contiguous recombination restores the target's expected query/head order before its batched output projection. PQ realizes the same mathematical separation by slicing one query and its mask column per subgraph. The question tested later is whether QB matches PQ's *observed numerical execution* without paying for the same subgraph structure.

3.3. Admission conditions are part of the method

The evaluated build admits one sequence/KV stream, non-flash attention, supported Qwen3 or dense Llama-family graph paths, compatible head dimensions, an integral GQA ratio, F32 query/mask values and F16 K/V. At most four queries are occupied in the current QB study. Attention bias, sinks and MLA value projections are excluded by the relevant source guards; other positional-bias schemes or unsupported strides require separate handling and validation. The mask may contain padded columns, but its first q logical query columns and original key extent must be represented correctly.

These conditions preserve distinct meanings for sequence ownership, query indexing and head grouping. Additional streams need their own batch coordinate; biases and sinks need their own broadcasting rules. The source guards enforce the supported feature set, and each backend validates operator admission. Wider-geometry and fused-attention paths have separate numerical and execution contracts (Supplement S1).

3.4. Phase binding and graph-cache consistency

The transformation is selected by a process-fixed four-bit mask: target prefill, target continuation, draft prefill and draft continuation have bits 1, 2, 4 and 8. Mask 3 enables the two target phases. A scoped guard derives the active flag from the current role and phase, places that flag in graph parameters, and restores its prior value on exit. New requests reset the phase to prefill. Crucially, this guard executes before the observation-disabled fast path returns to native decoding; turning off a diagnostic callback must not disable the repair.

Graph caching creates a second obligation. Two calls can share nominal capacity and tensor extents yet require different attention layouts. The active-layout flag therefore participates in the reuse check. A mismatch rejects reuse and requires an appropriate graph. The original campaigns fix the implementation choice (QB, PQ or GS) within each process. The later matched native bridge uses separate contexts for each layout within a process (Section 6.1). Both keep the implementation fixed within a context; they do not validate dynamic PQ/QB switching under one cached graph. Table 1 summarizes the dispatch and reuse checks.

The surrounding operators remain batched. Complete-output qualification therefore tests the transformation within its numerical stack, including state formed upstream. The Vulkan projection interventions in Section 7 locate this producer-side dependency.

Table 1: Phase-scoped dispatch and reuse obligations.

1	On request reset, establish target/draft ownership and set phase to prefill.
2	Compute the role/phase bit and save the previous scope flag.
3	Set the active flag before both observed and unobserved decode paths.
4	If cached topology's layout flag differs, reject reuse.
5	For an active supported attention graph, apply Q/mask views, native QK–softmax–AV and ordered recombination.
6	Otherwise retain the declared untransformed graph or reject an unsupported active case.
7	Complete native acceptance/state updates and restore the prior scope flag.

3.5. Selecting the active phases

A prior four-switch experiment tested all 16 masks on four exposed HumanEval inputs at $k = 3$, UB4 and cap128. Masks 3, 7, 11 and 15 passed both repetitions; mask3 was inclusion-minimal within that finite mask family. It was then fixed for the later model-family tests. The complete pattern and the separate mask3/mask15 cost comparison appear in Supplement S1; that comparison did not pass its practical speed threshold.

4. Experimental design

4.1. Systems, panels and distinct evidence roles

The implementation uses pinned llama.cpp/GGML sources [10, 9]. CUDA experiments run on an RTX 5070 Ti Laptop GPU with Q8_0 target/draft pairs: Qwen3-8B/0.6B [11], SmolLM2-1.7B-Instruct/135M-Instruct, and Llama-3.1-8B-Instruct/3.2-1B-Instruct. The models share a backend but include different head dimensions and attention-head mappings. Weight files and runtime libraries are byte-pinned, with exact versions recorded in the ancillary source manifest.

The ordinary CUDA reference uses physical UB1. Repaired execution uses target caller UB4, draft UB1, $k = 3$, F16 KV, non-flash attention and context size 1024. The requested logical batch of 2048 is clamped to 1024; the frontend uses 24 host threads. One request is active at a time and prompt-cache reuse is disabled. The original Vulkan state studies instead use an Arc B570 and Llama-3.1-8B-Instruct Q6_K, with F16 KV, context 1024 and ten host threads. They are state and portability experiments, not a controlled comparison of GPU speed.

The core CUDA panels use sanitized MBPP and GSM8K inputs [12, 13]; HumanEval supplies engineering and Vulkan service probes [14]. The new-input transfer panel adds CodeContests programming tasks [7]. Outcomes are response preservation and runtime. Table 2 maps each panel to its research question and distinguishes repeated execution from new-input validation.

The 24-input output-confirmation panel was frozen after exclusion against accessible project execution history. The repair comparison and E2 reuse 12 exposed texts, six from

Table 2: Principal evidence groups. Counts include the indicated repeats, not independent tasks. Exact sources, failure records and access scope are indexed in Supplement S10.

Study	Design	Question answered
CUDA output confirmation	24 project-unused inputs per family; 5 arms; 2 sessions; 5 repeats; 3,600 requests	Complete-output qualification of a frozen repair and ordinary baselines
CUDA repair comparison	12 exposed inputs per family; 7 arms; 2 sessions; 4 repeats; 2,016 formal requests	PQ/QB/GS costs and matched draft increment
CUDA E2 replication	Same 12-input panel; 6 separated sessions; 324 fresh workers; 1,620 requests including 324 warmups	Independent-process PQ/QB comparison across two dates; retrospective local draft-on/off analysis
CUDA E1 replay	3 families × 3 layers × 2 phases × 4 query widths; 216 model requests and 432 replays	Same-input QK, softmax and AV correspondence
Vulkan state studies	Prefill partitioning, admitted whole-state crosses, and 32 controls plus 160 subset requests	State formation and component substitution
Vulkan port and source audit	Stopped natural pilot; 432 synthetic CPU fixtures and frozen support-predicate analysis	Endpoint counterexamples and backend admission
Later mapping studies	RTX replay/padding controls and independent B570 representation transactions	Mapping admissibility under a strong software baseline
Fresh-context service panels	Four exposed inputs and four process sessions per platform; separate references and policies	Real two-query execution, layout cost and draft economics
Matched CUDA deployment	12 exposed Qwen3 inputs; $k = 1, 3$ by resident/fresh; 6 process sessions on one date; 864 measurements and 72 warmups	Layout benefit under matched draft length and request lifecycle
New-input CUDA transfer	12 inputs planned per family; 33/36 family/input cells reached; 4 sessions per qualified family; 384 measurements and 384 warmups	Qwen3/SmolLM2 qualification and cost on all 12 inputs each; Llama stops at a reference counterexample

each domain; the matched deployment study retains the Qwen3 panel. E1 selects one exposed prompt per family before capture. The new-input transfer panel is frozen separately after exclusion against project history. Vulkan state interventions use two known divergent inputs, two stable controls and separately labelled synthetic prefixes. Each panel supports the claim listed in Table 2.

4.2. Matched alternatives and complete-request timing

The seven repair-comparison arms are R, M_{QB}, C_{QB}, M_{PQ}, C_{PQ}, M_{GS} and C_{GS}. M is target-only; C adds the same draft protocol. GS physically serializes active target rows without changing the registered draft or stopping rule. PQ and QB retain surrounding batching. The original campaign uses arm-specific workers and reverses their order in its second session. E2 retains R and the four PQ/QB arms, using two serial-reference endpoints for each task/session.

The paid CUDA interval is a complete nonstreaming resident-process HTTP request, including encoding/submission, generation, response parsing and endpoint validation. Model loading, artifact writes, hashes and registered warmups are outside it. The counts channel adds no heavy tensor readback or observation synchronization. Diagnostic times are separate and are not subtracted from request measurements.

For task i, session s and arms a, b, let $t_{i,s,a}$ be the registered request measurement, or the within-session median where repetitions are specified. We use equal task and session log weights:

$$S_{a/b} = \exp\left(\frac{1}{|\mathcal{S}|}\sum_{s}\frac{1}{|X|}\sum_{i}\log\frac{t_{i,s,a}}{t_{i,s,b}}\right). \tag{7}$$

Values above one favor the denominator. Full required pairs must qualify; missing and divergent records are retained without imputation. On a common paired panel, ratio products obey algebraic identities, but they do not decompose operator time.

E2 uses fresh processes, short PQ/QB blocks and counterbalanced order. Six sessions span 24.45 hours from first to last start, with at least one hour between a completion and the next start. The primary interval is the registered pointwise 95% percentile bootstrap of six session-level log contrasts, using 10,000 resamples. Its sampling unit is the session and its target is the fixed 12-input panel on one host. The practical ratio threshold is 1.02: for PQ/QB, this corresponds to a QB latency reduction of $1 - 1/1.02 \approx 1.96\%$. The later service studies register a 2% reduction relative to their PQ control mean. No request-level bootstrap, outlier trimming or significance-dependent stopping is used.

A secondary sensitivity analysis applies a two-sided Student t_5 interval to the same six session mean log contrasts

and recomputes the estimate after omitting each session. The t interval assumes independent, approximately normal session log means. It checks dependence on the interval method and individual sessions while retaining the registered bootstrap as primary.

The retrospective E2 draft-on/off analysis pairs the four local arms within each family/task/session. It applies the same equal log weights to M_L/C_L and reports exploratory pointwise t_5 intervals across six session means. These contrasts were not preregistered, and their intervals provide no simultaneous-coverage guarantee. They use no additional requests and leave the registered PQ/QB estimands and intervals unchanged.

4.3. Numerical and state controls

E1 freezes actual Q/K/V/mask bytes, shapes, strides, head maps and parameters. Native MQ capture/replay correspondence, within-arm repeatability and capture-on/off endpoint equality precede PQ/QB comparison. Each of 72 fresh workers runs a counts warmup, a heavy capture and a counts observer control; MQ, PQ and QB are then replayed twice per context. A failed control withholds the corresponding numerical claim.

Vulkan interventions reconstruct a declared continuation boundary: complete sequence KV and positions, initial full FP32 logits consumed by the native greedy sampler, an empty generated prefix and correct pending-token ownership. Sixteen own-path shams precede whole-KV crosses. Subset tests retain recipient initial logits and verify actual imported bytes. Subsequent four-cell analyses compare full logits only at a shared consumed prefix.

The service and mapping studies specify their own reference and cost windows (Sections 6.2 and 7.6); the matched deployment design changes k and lifecycle within a common runtime (Section 6.1). Campaign identities, amendments and execution dispositions are indexed in the supplement. The resident HTTP comparisons retain their stated $k = 3$ policy, timer and task weights.

5. Repair scope and query-layout cost

5.1. Repair scope changes draft profitability

The seven-arm campaign compares repair scope and drafting on the same 12-input panel. All seven arms qualify 96/96 complete endpoints per family under the declared reference. Figure 2 separates local representation cost, global serialization cost and the draft increment under all three repair organizations. These estimates retain the original two-session protocol, separately from E2.

Its speculative GS/QB ratios are 2.144, 1.463 and 2.176; every task favors QB. These differences reflect the value of constraining attention locally while retaining batched projection and feed-forward work. The PQ/QB ratios, 1.044, 1.068 and 1.022, measure the representation choice within local repair. These common-panel estimates distinguish repair scope from layout cost; E2 supplies a separately scheduled PQ/QB comparison.

A retrospective audit of all 2,016 formal responses forms 288 matched seven-arm audit groups. All 1,728 within-path GS/PQ/QB pairings agree in recorded proposals, acceptance, logical rows, rollback and caller-width histograms; complete native token IDs and recorded EOG flags match the paired R endpoints. GS changes physical batching as intended; PQ/QB agree in every workload field. This holds recorded logical work fixed without apportioning operator time (Supplement S2.3).

Adding the draft to QB is profitable for Qwen3 and Llama but not SmolLM2: $M_{\mathrm{QB}}/C_{\mathrm{QB}}$ is 1.144, 0.634 and 1.117. SmolLM2 takes about 1.58 times its own draft-free request duration despite QB being cheaper than PQ. Across the common seven-arm panel, draft value under another layout L follows

$$D_L = \frac{S_{M_L/M_{\mathrm{QB}}}}{S_{C_L/C_{\mathrm{QB}}}} D_{\mathrm{QB}}, \qquad D_{\mathrm{QB}} = S_{M_{\mathrm{QB}}/C_{\mathrm{QB}}}. \tag{8}$$

The derived GS/PQ/QB values are 0.710/1.125/1.144 for Qwen3, 0.592/0.616/0.634 for SmolLM2 and 0.724/1.107/1.117 for Llama. Thus the repair policy changes the sign of the observed draft decision for two families. Both PQ and QB cross the profitability boundary; their direct comparison measures QB's additional layout benefit.

The relative change in draft value summarizes this interaction:

$$\Gamma_{\mathrm{QB}\leftarrow\mathrm{GS}} = \frac{D_{\mathrm{QB}}}{D_{\mathrm{GS}}} = \frac{S_{C_{\mathrm{GS}}/C_{\mathrm{QB}}}}{S_{M_{\mathrm{GS}}/M_{\mathrm{QB}}}}. \tag{9}$$

It is 1.612, 1.071 and 1.542 for the three families, respectively. These are retrospective ratio-of-ratios summaries from the reported common-panel estimates, with the original weights and timer. They quantify how repair policy changes the draft increment at complete-request level. The draft's incremental cost must therefore be evaluated under the repair policy being considered.

Reference-preserving batching also has value without drafting. Target-only QB lowers cost relative to the specified serial reference: R/M_{QB} is 1.334051, 1.369890 and 1.426161, with every task favoring QB on this panel. For SmolLM2, the measured choice is therefore to retain target-only QB batching and omit the draft. These complete-request gains include prompt prefill; the target-only continuation uses a common one-query path. They establish neither a single-token decoding speedup nor a measured decomposition by phase.

A historical fixed-UB4 experiment provides a related counterexample: increasing k from 3 to 7 preserved outputs but reduced request speed to 0.890 of the smaller-draft path. Separate work captures recorded fewer target callers and more physical micrographs. Acceptance and caller reduction alone therefore cannot establish request acceleration. The width failures and negative cost results that constrain this conclusion remain in Supplement S1.

5.2. Draft decisions across the six E2 sessions

E2 records target-only and speculative requests under each local layout. We retrospectively pair its 864 non-warmup local-arm measurements into 216 family/task/session quartets: M_{PQ}, M_{QB}, C_{PQ} and C_{QB}. The contrast $D_L = S_{M_L/C_L}$ uses the weights in Equation 7.

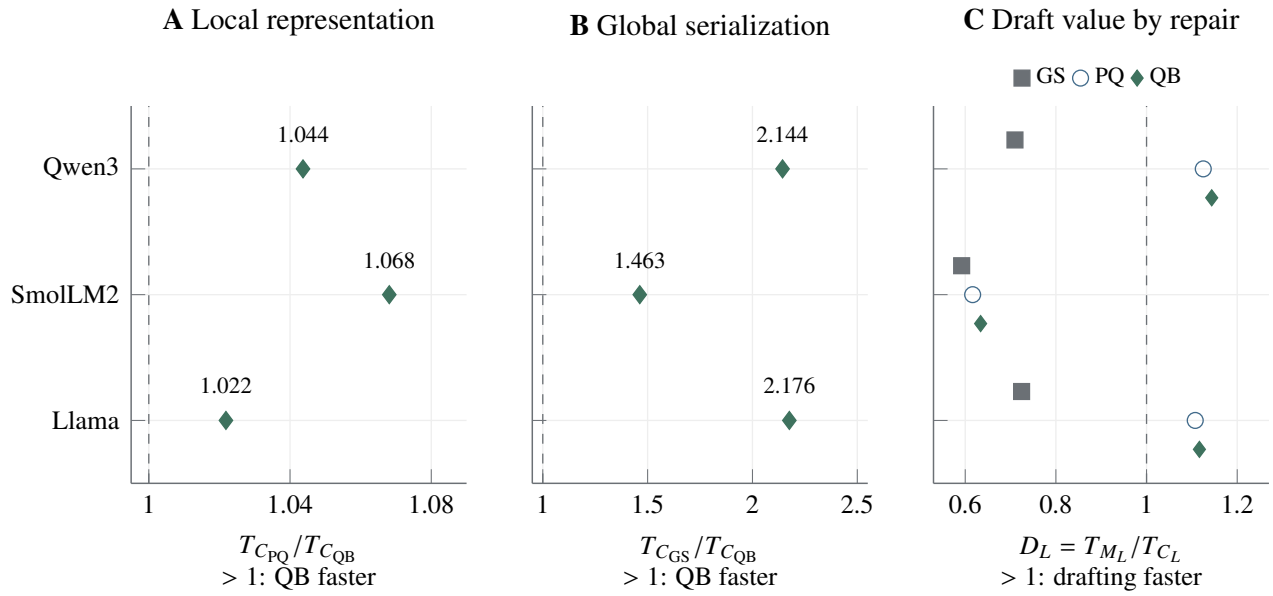


Figure 2: Matched repair and drafting choices in the original seven-arm campaign, separate from E2. A/B: speculative PQ and GS costs relative to QB. C: draft value under each repair organization, $D_L = T_{M_L}/T_{C_L}$; values above one favor drafting. Qwen3 and Llama cross parity under both local repairs, while SmolLM2 remains below it. GS/PQ values in C are reconstructed by Equation 8 from the common-panel ratios. All points are descriptive estimates; exact source ratios and every regression are retained in Supplement S2.

Table 3: Retrospective E2 draft values on the fixed 12-input panel: $D_L = T_{M_L}/T_{C_L} > 1$ favors drafting. QB intervals are exploratory pointwise 95% t_5 intervals over six session log means. E2 has no GS arm.

Family	D_{PQ}	D_{QB}	QB interval
Qwen3	1.072	1.084	[1.059, 1.109]
SmolLM2	0.576	0.597	[0.582, 0.612]
Llama	1.091	1.105	[1.097, 1.113]

Table 4: Absolute E2 request costs (ms), with equal task and session weights in the geometric mean. Registered warmups are excluded; the paid interval includes submission, full-response parsing and endpoint validation.

Family	M_{PQ}	M_{QB}	C_{PQ}	C_{QB}
Qwen3	2130.4	2087.5	1987.2	1926.3
SmolLM2	713.1	693.8	1238.8	1162.3
Llama	2574.7	2535.1	2360.9	2294.2

Table 3 retains all three local-policy directions: drafting benefits Qwen3 and Llama, while SmolLM2 favors target-only execution. Every one of the six session estimates has the corresponding direction under both PQ and QB. Under QB, drafting reduces complete-request latency by 7.72% and 9.51% for Qwen3 and Llama; it increases SmolLM2 latency by 67.52%. The magnitudes differ from the original two-session campaign, while the policy decisions agree.

The exploratory pointwise t_5 intervals use sessions as sampling units; the registered PQ/QB analysis is unchanged. Pairing separately collected target-only and speculative blocks does not remove within-session drift. Without a GS arm, E2 checks the local draft decision rather than independently replicating the full repair-scope interaction. Complete estimates and the PQ-to-QB change in draft value appear in Supplement S3.3.

5.3. QB's independent implementation benefit

The registered E2 comparison isolates representation cost within local repair. It completes 1,620 requests in 324 fresh workers, including 324 warmups and 432 serial-reference measurements. There are no request failures, output mismatches, workload errors or heavy-observer flags. The six-session speculative ratios C_{PQ}/C_{QB} are 1.031614, 1.065803 and 1.029114, corresponding to 3.06%, 6.17% and 2.83% lower QB request latency. Figure 3 shows both paths and their session-level uncertainty.

Table 4 gives absolute costs from the source-verified requests, using the same task/session log weights.

All 36 family/path/session contrasts favor QB; 9 of the 72 family/path/task aggregates regress. The primary pointwise speculative intervals are [1.025393,1.037926], [1.063387,1.068399] and [1.020126,1.039149], each above the predeclared 1.02 threshold under the registered percentile method. Target-only effects are smaller: Llama's estimate is 1.015593, below the threshold, while Qwen3's interval overlaps it.

The secondary t_5 intervals remain above 1.02 for speculative Qwen3 and SmolLM2. Llama's interval is [1.015387,1.043026], above parity but crossing 1.02; its six leave-one-session-out estimates range from 1.024960 to 1.032333. QB's advantage is stable in these checks. Whether the Llama effect exceeds the practical threshold depends on the interval method. Supplement S3 reports both analyses and all task regressions.

E2 establishes endpoint and recorded-counter correspondence across all 216 speculative PQ/QB task/session pairs: acceptance, rollback and logical-workload counters agree. This supports an implementation-cost comparison under matched recorded work. The original two-session estimates and their target-only direction reversals are reported in Supplement S2; E2's recorded-counter scope is detailed in Supplement S3.

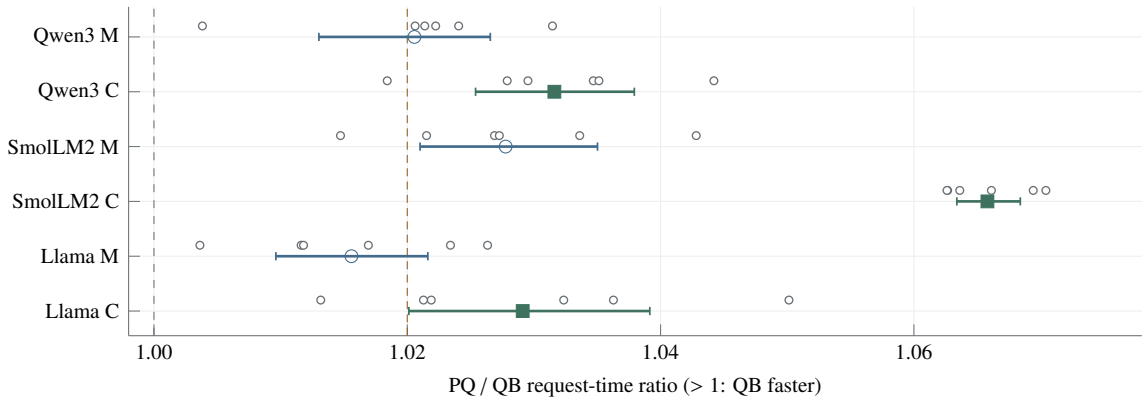


Figure 3: Independent-block E2 replication. Small marks are the six session estimates; large markers and horizontal bars are the equal-weight estimate and its registered 95% six-session percentile interval. Dashed lines mark parity and the 1.02 practical threshold. *M* is target-only and *C* speculative. The fixed exposed panel and single host limit statistical scope.

Bitwise-equal contexts / 72

	QK	Softmax	AV
MQ vs PQ	18 / 72	18 / 72	18 / 72
MQ vs QB	18 / 72	18 / 72	18 / 72
PQ vs QB	72 / 72	72 / 72	72 / 72

18 single-query controls; 54 multi-query contexts

Figure 4: Completed 72-context CUDA replay. The 18 agreements between MQ and either repair are exactly the $q = 1$ common-path controls. The 54 multi-query cases first differ at QK. All PQ/QB stage outputs agree bitwise.

The recovered E2 records do not report actual CUDA Graph capture, replay or fallback counts. The request-cost estimate therefore identifies the implementation effect but does not isolate those mechanisms. The controlled QB/QH replay study in Section 7.6 has its own execution evidence and comparator.

5.4. Same-input correspondence across layers and phases

The initial 24-context replay established PQ/QB correspondence at sampled layer-0 attention boundaries. E1 expands the test to three layer positions, prefill and continuation, and $q = 1, 2, 3, 4$ in each family. All 72 contexts pass observation, native-MQ own-path and repeatability checks. PQ and QB are bitwise equal at QK, softmax and AV in every context, with zero maximum error and ULP distance (Figure 4).

The sampled target layers are 0/18/35 for Qwen3, 0/12/23 for SmolLM2 and 0/16/31 for Llama. Qwen3 and Llama use head dimension 128 with 32 query heads and 8 KV heads; SmolLM2 uses dimension 64 with 32/32 heads. Thus the observed correspondence spans both ordinary and grouped-query attention and more than one head dimension.

MQ agrees with either repaired path only in the 18 single-query controls. Across the other 54 contexts, its first difference is at QK; maximum finite absolute differences are 0.078583 at QK, 0.000917 at softmax and 0.010433 at AV. These are operator differences on identical operands, not model-logit bounds.

All snapshots have tensor key extent 256, including masked padding. The maximum number of visible keys is 109, 130 and 127 by family; prefill captures include early query positions with only one to four visible keys. The experiment covers layers, phases and query widths at these short visible contexts, using one exposed prompt per family. Complete coordinates and operator comparisons appear in Supplement S3.

6. Complete-service deployment cost

6.1. Matched deployment: k3 gains survive fresh contexts

A new matched study separates the effects of speculative depth and context lifecycle within one runtime [6]. It uses the original twelve exposed Qwen3 inputs, unchanged model/tokenizer bytes, context1024, target UB4, draft UB1 and cap128/native EOG. One executable supplies all four $k = 1, 3$ by resident/fresh cells and the PQ_A/PQ_B/QB arms. All 204 qualification requests preserve complete outputs and speculative state; corresponding PQ/QB decision vectors agree, and every output also matches the historical E2 reference.

Six independently loaded process sessions complete 864 measurements and 72 warmups. The order balances cells and arms within input/session blocks. Both models remain loaded. Resident requests clear their existing contexts; fresh requests create and destroy contexts inside the service timer. The shared timer includes prefill, generation, sampling, synchronization and result construction. Heavy capture and additional backend counters are disabled.

QB reduces service time by 6.26% at resident $k = 3$ and 5.36% at fresh $k = 3$ (Table 5). Both pass the registered practical screen; neither $k = 1$ cell passes. The $k = 3$ point reductions exceed their respective 4.125% and 4.388% input-aggregate A/A imbalances and favor QB in every session. The four primary intervals resample whole process sessions and use a Bonferroni adjustment; the twelve inputs remain a fixed panel.

The layout benefit therefore survives fresh-context service at $k = 3$. Secondary paired contrasts favor a stronger layout effect at $k = 3$ in both lifecycles, while each individual lifecycle contrast includes parity. Changing k changes the speculative transaction as well as query width. These

Table 5: Matched Qwen3 deployment bridge. Ratios compare QB with the arithmetic mean of identical PQ controls; aggregation gives equal weight to inputs and process sessions. Intervals use the registered Bonferroni-adjusted bootstrap of six process sessions, with nominal 95% family coverage across the four cells. The practical screen also requires reduction above the A/A threshold and improvement in every session.

k	Lifecycle	QB/dual-PQ [adjusted interval]	Practical gain
1	Resident	0.986617 [0.965043, 1.006956]	Not established
3	Resident	0.937404 [0.922603, 0.949354]	Established
1	Fresh	0.973534 [0.955904, 0.996512]	Not established
3	Fresh	0.946409 [0.940517, 0.953312]	Established

comparisons identify a deployment-dependent layout effect, not a pure width intervention or a CUDA Graph mechanism. The new driver also does not reconstruct every difference between the older HTTP E2 and fresh-service contracts.

6.2. *Historical k1 service controls*

The earlier RTX and B570 studies place the matched result in context. Both use Qwen3-8B/0.6B Q8_0, $k = 1$, F16 KV, non-flash attention and four exposed inputs. Their request formats, stopping windows, prefill policies and drivers differ. Each timer charges fresh-context creation, graph reservation, prefill, generation, synchronization and destruction, with model loading separate. Full contracts and all blocks appear in Supplement S8 [15, 16].

The paired layout estimator uses two identical PQ controls:

$$\bar{T}_{is,\mathrm{PQ}} = \tfrac{1}{2}(T_{is,\mathrm{PQ}_A} + T_{is,\mathrm{PQ}_B}), \qquad r_{is} = T_{is,\mathrm{QB}}/\bar{T}_{is,\mathrm{PQ}}. \quad (10)$$

Input/session ratios are combined with equal log weights. The finite-panel screen requires a reduction above both 2% and the largest input-aggregate A/A imbalance, with improvement in every session. Both original panels fail this practical layout screen: QB/dual-PQ is 0.995144 on B570 and 0.989284 on RTX. The corresponding point reductions are 0.486% and 1.072%, against A/A imbalances of 0.754% and 5.208%; two of four RTX sessions favor PQ.

The speculative policy has a different outcome. On RTX, QB/R=0.582325 and dual-PQ/R=0.588633, so PQ already obtains most of the 41.8% service-time reduction. On B570, QB/R=2.619763 under its separate two-model Vulkan contract. Its qualified R/PQ/QB requests match complete endpoints, 346 PQ/QB full-logit pairs and cycle traces, while costly draft work dominates the request. Changing the draft thread parameter produces DRAFT1/dual-baseline=1.003994 [17]. The later CPU-draft prefill-submission study gives SUBMIT/dual-baseline=0.999161 and also fails its practical screen [18].

These controls distinguish the speculative policy's value from QB's incremental layout value. The common-runtime study above tests draft length and lifecycle directly; the original panels retain their separate references and timers. Their complete qualification, phase intervals, warmups and execution records remain in Supplement S8.

7. Reference transfer across inputs and backends

7.1. *Transfer to previously unexposed inputs*

A separate study retains the three-family $k = 3$ resident HTTP contract and tests twelve previously unexposed natural inputs per family [7]. The frozen panel covers formatted prompt-length strata 64–192, 256–448 and 640–832 within context1024. Short inputs are GSM8K mathematics; longer inputs are CodeContests code. Length and domain therefore vary together.

Qwen3 and SmolLM2 pass all twelve inputs. Four sessions per family supply 384 measurements and 384 warmups in total. SmolLM2's 6.99% QB layout reduction passes the finite-panel practical screen, yet QB/R is 1.007007: the complete speculative request remains slightly slower than its serial reference. Qwen3's 3.22% reduction is smaller than its 4.78% A/A threshold. Layout value and profitable speculation remain separate questions.

Llama provides a reference-transfer counterexample. On CodeContests/valid/108, with 737 formatted prompt tokens, PQ and QB agree with each other in outputs, full decision vectors and causal trace, but both depart from deployed R at zero-based output index81. The remaining three Llama inputs and all Llama timing slots stay unexecuted. PQ/QB correspondence therefore does not establish reference preservation on a new input. The tested native-fast comparator also fails reference qualification in all three families; no qualified timing victory over that route is claimed. Supplement S9 gives accounting and source access.

7.2. *Qualify the producer state*

A local attention repair consumes state produced upstream. The B570 studies test whether prefill partitioning changes that state and the resulting single-query continuation. With target context UB512 and output capacity one, arm A submits the whole prompt; B splits it into 128 tokens and a remainder. Arm C uses UB128 with internal partitioning. All arms request final-position logits and then continue naturally.

For two exposed inputs of 161 and 133 prompt tokens, B reproduces C's complete endpoint, initial FP32 logits and sequence-KV bytes. A diverges at generated indices 47 and 81. Two longer controls have unequal A/B states but equal endpoints. A partition sweep yields several endpoint classes, including a third trajectory for the 133-token input split as 132 + 1: smaller chunks are not monotonically closer to the reference (Supplement S4).

The differences begin before attention: first-layer K/V projections differ for the 161-token input, and Q/K/V differ for the 133-token input. Fixed-operand replay qualifies 32 of 40 calls and reproduces projection sensitivity. Eight scale/MUL calls are excluded because an in-place consumer overwrote their captured operands. The partition

Table 6: Transfer to previously unexposed inputs within context1024. Each family has a registered 12-input panel. Qwen3 and SmolLM2 complete all inputs; the Llama reference failure stops that family before cost measurement. The layout screen uses a 2% floor, input-aggregate A/A and every-session direction; no population interval is assigned.

Family	Reference qualification	QB/dual-PQ (screen)	QB/R
Qwen3	12/12 inputs pass	0.967781 (not established)	0.560577
SmolLM2	12/12 inputs pass	0.930056 (pass)	1.007007
Llama	PQ and QB both depart from R	Not run	Not run

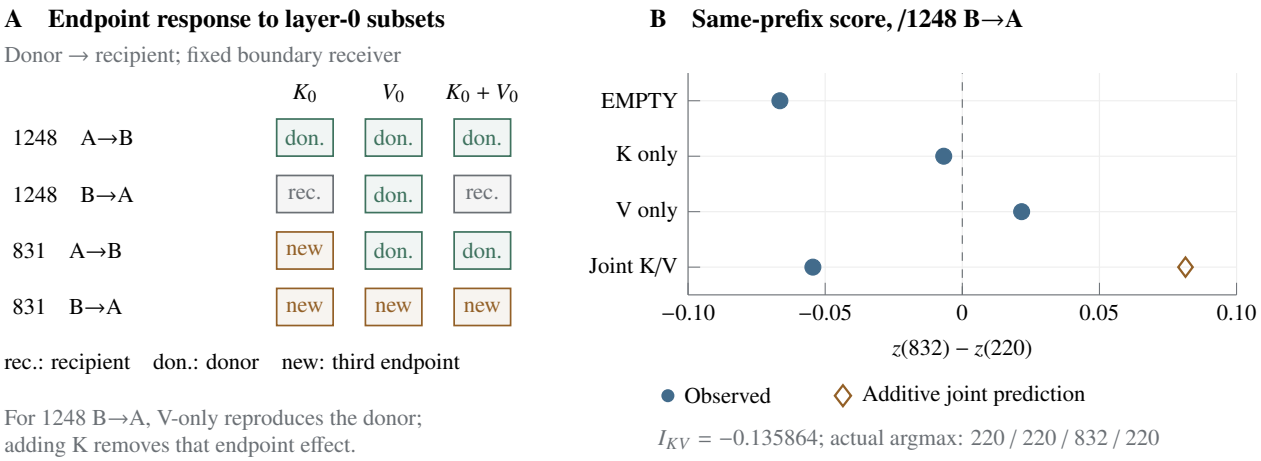


Figure 5: State composition at two levels. A: layer-0 endpoint classes; rec. retains the recipient, don. reproduces a distinguishable donor and new is a third trajectory. Each cell has two agreeing process repeats. B: full-logit comparison for /1248, B→A, at output 47 after a common consumed prefix. The hollow diamond is the additive prediction, not a fifth execution.

sweep is diagnostic because its driver body was authenticated after execution; the later replay has an authenticated admission record (Supplement S4).

A common UB512 receiver isolates the state link. It imports sequence KV and positions, samples the selected complete initial logits, and restores pending-token semantics. Sixteen own-path shams reproduce natural endpoints; 40 crosses then vary KV and logit sources. For the two affected tasks,

$$Y(KV_A, L_A) = Y(KV_A, L_B) = Y_A, \tag{11}$$
$$Y(KV_B, L_A) = Y(KV_B, L_B) = Y_B. \tag{12}$$

All initial vectors choose token 1271, making the logit swap a first-decision negative control. The suffix follows the KV source; the stable controls retain their endpoint despite unequal KV. Local consumer qualification alone cannot certify the producer state. Transfer requires qualification of both.

7.3. Endpoint agreement and state substitution

The subset study tests ten selectors on four exposed inputs in both donor directions. After 32 EMPTY/ALL controls, 80 coordinates each run in two independent processes, with agreement at every coordinate. Of the 160 requests, 30 reproduce a distinguishable donor endpoint, 54 produce a third trajectory and 76 retain the recipient endpoint despite changed bytes (Supplement S5).

A layer-0 K-only or V-only replacement selects 1/64 of total K/V value storage and can reproduce a donor endpoint. This sufficiency is nonmonotone. For /1248 B→A, layer-0 V reproduces the donor, but adding K removes that effect. In the reverse direction, layer-0 V works while replacing V at every layer does not.

Endpoint equality also fails as a cache-substitution criterion. Natural A/B states for /1011 and /1249 have equal endpoints but unequal KV. On /1011, 16 of 20 subset/direction coordinates produce a third trajectory; all 20 retain the endpoint on /1249. These interventions refute arbitrary component interchangeability; they do not establish natural reachability of mixed states.

To distinguish additive changes crossing an argmax threshold from a nonadditive score response, we compare full FP32 logits at the first divergence after a common consumed prefix. For fixed tokens (r, u), write $d_S = z_S(u) - z_S(r)$ and

$$I_{KV} = d_{KV} - d_K - d_V + d_\emptyset. \tag{13}$$

For /1248 B→A at output 47, $(r, u) = (220, 832)$. EMPTY/K/V/KV scores are −0.066515, −0.006771, +0.021633 and −0.054487. The additive prediction is +0.081377, opposite in sign to the observed joint score; $I_{KV} = -0.135864$. Full-vocabulary argmax tokens are 220/220/832/220. The result establishes a nonadditive response through subsequent computation, without localizing it to one kernel.

Supplement S5 records the 44 complete contrasts among 112 planned repeat-level contrasts, canonical token coordinates and missing vectors.

7.4. Qualify backend execution

The original Vulkan pilot combines reference-like or bulk prefill with MQ, PQ or QB continuation. Reference monitoring fails the declared sampling schedule: 48 of

49 attempted requests complete, one is interrupted and 31 coordinates remain unstarted. Speculative QB also changes the /831 and /1011 outputs. The pilot fails qualification; Supplement S6 retains its full record.

Independent address and scalar oracles pass 432 synthetic fixtures over 72 geometries. Yet the original $q = 2$ QB graph violates a necessary condition of the frozen Vulkan matrix-multiplication backend:

$$b_3(\mathrm{src}_0) = b_3(\mathrm{src}_1), \tag{14}$$

where b_3 is the fourth tensor extent. QK pairs K at extent 1 with lifted Q at extent 2; AV pairs V at extent 1 with probabilities at extent 2. Both are rejected. MQ and each PQ slice satisfy this check with (1, 1).

The frozen backend rejects these lifted QK and AV descriptors because their fourth source extents differ. Actual placement and fallback transfers were not observed in the failed pilot; the later port addresses the source-level admission requirement directly.

7.5. Preserving the AV path in a scoped GQA port

A scoped port targets F16/F32, $d = 128$, 32-query-head/8-KV-head layouts, physical key extents 256/512 and $q = 2, 4$. Its packed route matches PQ at QK and softmax but differs in 7,160 of 8,192 AV values at key512/$q = 2$, with maximum absolute difference 1.22236×10^{-9}. That route fails the exact reference [19].

The successful route reuses the existing noncontiguous AV shader with zero V batch stride, real probability/output strides and independent query workgroups. All six key256/512 by $q = 1, 2, 4$ coordinates match byte-for-byte at QK, softmax, AV and final output; all 20 earlier PQ stage files remain unchanged. The port retains per-output evaluation while exposing parallel work across queries. Section 6.2 qualifies this fixed-layout port in real requests; the earlier Llama Q6_K pilot remains failed.

7.6. Representation and execution-mode boundaries

A local representation must also satisfy the reference after backend path selection and repay its conversion costs. Query-to-head mapping (QH) places independent queries in a head-like dimension and restores native output order. Balanced operator studies compare it with two identical QB controls under fixed inputs and execution windows [20, 21].

On the RTX core panel, QB_A/QH=0.900826. A frozen selector predicts all twelve held-out synthetic mean orderings, but its charged lookup raises the offline selector/fixed-QB estimate to 1.010298. CUDA Graph ON/OFF intervention reduces QB completed-graph cost to 0.615–0.849 of OFF and narrows four QH-minus-QB gaps from 12.056–12.975 to 0.735–1.627 μs. These measurements establish execution-mode sensitivity of QB versus QH; the PQ/QB request comparison has a different comparator and lacks corresponding capture/replay counts.

Reference admission can exclude an otherwise faster execution path. RTX physical key extents 1023/1024/1025 select 8/3/8 kernels. Padding the same logical length-1023 operands and mask to extent1024 reaches three kernels on the first call but fails all four reference qualifications. Padding to1025 qualifies, retains eight kernels and costs about 1.5% more than native QB (Figure 6).

Charging copy/transpose, upload, completed graph execution and ordinary-output readback removes the apparent compact-QH advantage. RTX compact-QH/QB_A ratios are 1.002216 and 1.001598 at $q = 2, 4$, and neither passes the practical screen against both QB controls. The separate B570 transaction gives 0.992244 [0.979908,1.004691], also below the evidence required for a practical improvement. Fixed QB remains the selected baseline in these domains. Input-lifetime controls, continuous timer boundaries and all alternative mappings are detailed in Supplement S7.

8. Selecting a reference-qualified implementation

Repair scope, local representation and draft policy jointly determine request cost. For a declared workload X and service contract, let $\mathcal{A}_\rho(X)$ contain the evaluated candidate combinations that pass reference qualification. The selection objective is

$$a^\star = \arg\min_{a \in \mathcal{A}_\rho(X)} \widehat{T}_a(X), \tag{15}$$

where $\widehat{T}_a$ is the observed complete-request cost estimate, using the same estimator and weights across candidates. Table 7 separates the conditional comparisons needed to evaluate this joint choice. Their effects need not be separable; sequentially choosing each dimension need not find the cheapest combination.

Repair scope is a scheduling decision. The matched seven-arm panel shows that local repair changes the sign of the draft increment for two families. Both PQ and QB retain batched projections and feed-forward work; QB reduces the representation cost within local repair. E2 retains the local draft decisions across all six sessions. Target-only batching supplies a useful endpoint when the draft adds cost, as SmolLM2 demonstrates.

Numerical organization belongs in runtime interfaces. Paired Q/mask views carry query and head semantics; the phase flag and layout-sensitive cache check preserve those choices during graph reuse. The scoped Vulkan port retains the noncontiguous AV shader while exposing independent query workgroups [19]. Producer-state qualification completes this interface: endpoint-equivalent states can contain noninterchangeable KV components. Together these results identify the state, layout and evaluation properties a portable implementation must preserve.

Optimize the complete service transaction. The matched Qwen3 study establishes practical QB benefits at $k = 3$ with resident and fresh contexts [6]. The original $k = 1$ panels, compact-QH transactions and B570 prefill-submission study identify cases where reduced local work does not pass a complete-cost screen. The implementation choice therefore uses measured request cost under the intended draft length, backend and lifecycle.

These control points complement batch-invariant kernels [3], reproducible accumulation [22] and phase-aware

A Fixed logical length 1023: RTX

Physical extent	Qualification	Kernels	Cost / native A
1023 (native)	4/4 pass	8	1.000000
1024 (padded)	4/4 fail	3*	—
1025 (padded)	4/4 pass	8	1.014783

*First-call trace only; the failed path has no qualified clean cost. Logical inputs and original mask columns are unchanged.

B Complete Q-representation transactions

Panel	Comparator	Compact-QH ratio
RTX $q = 2$	QB_A / QB_B	1.002216 / 1.008295
RTX $q = 4$	QB_A / QB_B	1.001598 / 1.004167
B570 $q = 2, 4$	dual-QB	0.992244

B570 adjusted interval: [0.979908, 1.004691]. All tested practical-effect screens remain unresolved; no QH advantage is established.

Figure 6: Reference and transaction constraints on representation choice. A: padding the same logical length-1023 input to extent1024 selects a three-kernel path but fails all four reference checks. B: complete Q-representation transactions, with conversion and transfer charged. RTX and B570 use distinct geometries and execution modes; the B570 interval is descriptive and family-adjusted.

Table 7: Evidence for joint offline policy selection under a declared reference, workload and service contract. Each cost comparison is conditional on the other policy dimensions.

Choice	Evidence required	Resulting decision
Reference admission	Supported layout, producer state and complete-output checks against R	Admit passing workload/configuration pairs under the contract in Section 2.
Draft on or off	Target-only and speculative requests with the same repair, reference and timer	Use the cheaper qualified policy; Qwen3/Llama favor drafting under both local repairs, while SmolLM2 favors target-only execution in the original panel and the retrospective E2 analysis.
PQ or QB	Paired complete-request cost with A/A and session controls	Select QB where its incremental benefit passes the declared screen; matched Qwen3 $k = 3$ passes under both lifecycles, while $k = 1$ remains unresolved.

tiled execution [23]. They also give a concrete hardware-design requirement: preserve the declared per-query evaluation, including matrix-instruction behavior beyond dtype [24], while exposing independent work across queries. Hardware performance would require an equal-resource comparison on the request-critical path; the archived same-policy AV model supplies an evaluation-parity control [9].

9. Related work and evidence scope

9.1. Where numerical consistency is enforced

Query-axis parallelism is established attention structure. FlashAttention uses tiled attention to reduce memory traffic [25]; FlashAttention-2 distributes query rows across thread blocks and warps while sharing K/V [26]. These methods optimize fused evaluation and work partitioning. QB instead expresses independent queries through views of existing non-flash operators and integrates their one-query organization with phase selection, graph reuse and speculative state. Its qualification target is the output of a specified deployed numerical stack.

Table 8 compares the closest control points. Batch-invariant inference stabilizes reductions in normalization, matrix multiplication and attention [3]. CoRun combines isolated prefill, fixed-shape decode and selective kernel constraints [4]. LLM-42 runs a fast path, verifies fixed-shape windows and repairs the corresponding KV entries [5]. These systems define reproducible execution contracts; matching an already deployed quantized reference is a separate requirement.

QB's method is the integrated graph/runtime transformation. It reuses attention operators while changing query/mask representation, preserving head semantics and binding the layout to phases and graph identity. The Vulkan AV-preserving port shows that backend admission can still require an implementation change. Local numerical correspondence and stateful endpoint qualification test different parts of this contract.

Other approaches act at different boundaries. LayerCast separates storage and computation precision [27]; TBIK aligns reductions across tensor-parallel configurations [28]; MarginGate selects verification using decision margins [29]. Vosti specifies engine/kernel behavior across batching, chunking and cache changes and proves guarantees under its assumptions [30]. Our logical mapping proof, runtime guards, operator replay and endpoint tests supply finite qualification against the specified reference.

9.2. State and request cost

Stage-Replay demonstrates suffix sensitivity to KV transplantation [31], and cache-dependent divergence is reported in quantized serving [32]. Our fixed-precision interventions add component-substitution counterexamples and common-prefix score comparisons. Activation-patching studies establish the importance of intervention metrics and nonadditive component effects [33, 34]; our four-cell score contrast tests nonadditivity beyond an endpoint pattern caused only by thresholding.

Ragged-batch synchronization errors supply a separate source of speculative failures [35]. We check semantic indexing, finite-precision correspondence and backend admission separately. Work on verification cost and actual execution behavior [36, 37] also motivates complete-request measurement. Our matched comparisons add the repair–draft interaction and isolate QB's incremental cost relative to PQ.

Table 8: Numerical-consistency control points. Performance comparisons in this paper use GS and PQ in the same runtime.

Approach	Consistency target	Constraint and arithmetic	State and verification
Batch-invariant kernels [3]	Each token's result is invariant to batching and slicing	Stable reduction/tiling rules in kernels, including attention split size	Consistent KV layout before attention; no added verify–rollback loop
CoRun [4]	Stable token sequence across co-runners, batch positions and arrival orders in a fixed stack	Isolated prefill and fixed-shape CUDA-Graph decode; disable decode Split-KV and constrain collectives	Request-local cache and RNG state; shape control supplies the execution contract
LLM-42 [5]	Tokens agree with a fixed-shape verification schedule	Dynamic fast path plus fixed-shape replay; mostly existing kernels	Verify–rollback and replacement of decoded KV entries by verified entries
Phase-scoped QB	Complete output agrees with the specified ordinary reference on the qualified panel	Q/mask views preserve local one-query attention; surrounding computation remains batched	Phase/layout-aware reuse, producer-state qualification and native speculative reconciliation

9.3. Scope of the empirical results

The CUDA request results cover one sequence, non-flash attention, $q \leq 4$, Q8_0 weights, F16 KV and context1024. E2 estimates variation across six sessions on one host for a fixed 12-input panel; E1 samples one prompt per family at physical key extent 256 and short visible contexts. The new-input study extends formatted prompts to 811 tokens and observed continuation visibility to 938 keys. Its short mathematical and longer programming inputs vary domain and length together [7]. The Llama counterexample in Section 7.1 bounds reference transfer. All three families reject the tested native-fast comparator, so qualified cost comparisons remain against the declared serial and repaired paths.

The Vulkan state interventions concern four exposed inputs and finite greedy continuations. The stopped Llama Q6_K pilot remains unqualified. The Qwen3 Q8_0 port has its own reference and runtime; native replay covers four complete requests, a real layer-0/key512/$q = 2$ fixture and its first-query $q = 1$ slice [8]. The historical $k = 1$ service and prefill-submission screens, including the latter's incomplete helper-process accounting, are retained in Supplement S8.

The matched Qwen3 study estimates a joint draft-length/lifecycle contrast on twelve exposed inputs. Its six process sessions occur on one date; the four cluster-bootstrap intervals have nominal family coverage and quantify process-session variation on that host [6]. Individual lifecycle contrasts include parity. The service estimates include the full declared request window, while CUDA Graph capture/replay mechanisms and the remaining differences between historical drivers are unidentified. Contexts beyond 1024, concurrent throughput, energy and hardware efficiency are unmeasured. Native rebuild/replay establishes source-host execution; external access to bulk tensors and pinned numerical dependencies is specified separately from saved-data reconstruction and CPU checks in Supplement S10.

10. Conclusion

Repair scope is a system-level choice that changes the value of drafting. On the matched CUDA panel, attention-local repair makes drafting profitable for Qwen3 and Llama, where global serialization makes it costly. The retrospective E2 draft-on/off analysis retains these local-policy directions across six sessions, including SmolLM2's preference for target-only execution. Phase-scoped QB implements local repair through paired query/mask views, shared K/V, phase dispatch and layout-sensitive graph reuse. It lowers speculative resident-request latency over PQ by 2.83–6.17% across the three model families. Target-only batching also preserves the reference at lower cost when drafting is unprofitable.

Reference qualification covers the complete request and its producer state, query/head mapping and backend evaluation. The new-input Llama counterexample, in which PQ and QB agree but both depart from the deployed reference, demonstrates why correspondence between local implementations is insufficient. These interfaces define what must be checked when transferring the repair.

The matched Qwen3 study establishes QB's $k = 3$ benefit over the complete service lifecycle: latency falls by 6.3% with resident contexts and 5.4% with fresh contexts; neither $k = 1$ cell clears the practical screen. The resulting decision is to compare qualified combinations of repair scope, representation and draft policy under the intended service contract. Their complete-request costs determine which combination to use.

Data and code availability

The manuscript package includes reported data, analysis scripts, editable figures and CPU interface tests. The companion repository provides native source kits, real fixtures and complete request-reference bundles through authenticated access. RTX rebuilds the core/frontend with pinned numerical binaries; B570 rebuilds its numerical libraries and shaders. Exact model acquisition and runtime preparation are documented. Supplement S10 and `REPRODUCTION_ACCESS.md` identify the fixed materials, download checks and execution requirements.

Declaration of AI-assisted work

OpenAI ChatGPT and Codex assisted implementation, analysis code, figures and writing. Measurements come from recorded executions. The author is responsible for verifying the results, interpretations and references and approving the final text.

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Supplementary material

Phase-Scoped Query Layouts for Reference-Preserving Speculative Decoding

Complete comparison tables and reproduction details. Section, figure, table and equation numbers use an S prefix; pages and citations continue from the article. Development history and detailed audit logs are indexed in the ancillary files.

S1: implementation S2–S3: CUDA evidence S4–S6: Vulkan evidence S7: mapping S8: service S9: deployment and transfer S10: reproduction

S1. Implementation and reference contracts

The CUDA target/draft pairs use Q8_0 conversions and F16 KV; the original B570 state-study target uses Q6_K and the historical draft Q4_K_M. Model families, converted files and reference recipes remain distinct between campaigns. Native endpoint checks retain IDs, EOG flags, length, last token and stop reason. Older first-EOG or retokenized-text comparisons have their original scope in the ancillary history; they do not qualify newer full-array comparisons.

S1.1. Implementation checks

Query views must fit the allocated source span, preserve alignment and nonoverlapping logical coordinates, and maintain the original mask column and GQA mapping. K/V ownership and the output permutation are checked explicitly. The role/phase flag enters graph parameters before the observation-disabled branch, participates in topology reuse and resets on a new request. PQ slices identical operands and masks; GS actually changes physical target microbatching. Neither baseline changes the draft, reconciliation or stopping rule to create a cost difference.

S1.2. Complete phase-selection experiment

The four-bit experiment tests four HumanEval probes, two repeats, $k = 3$, UB4 and cap128: 128 native requests. Table S1 preserves every mask. Mask3 is inclusion-minimal within this tested family and is fixed in the later campaigns. A separate 19-task/six-repeat comparison against all-phase mask15 gave only 1.003569 generation speed, below its registered 1.03 threshold, with an interval including parity. Removing constraints was not itself an established efficiency improvement.

S1.3. Width and runtime boundaries

The earlier joint (k, U) sweep tested $(1, 2), (3, 4), (5, 6), (7, 8)$ at caps 128/256/512 on 11 inputs. MBPP229 failed in the two wider geometries; a separate HumanEval18 failure retained its own denominator. The maximum observed reference length was 261, so requesting cap512 did not establish a 512-token complete trajectory. In a fixed-UB4 study, widening k to 7 preserved endpoints but yielded generation/request ratios of 0.829442/0.889868 versus $k = 3$. Draft proposals increased from 3,528 to 5,392 while accepted draft tokens increased from 2,772 to 3,192. Different captured sessions showed caller counts falling from 305 to 199 and micrographs rising from 305 to 397; these are not additive component timings.

Table S1: Historical finite four-switch pattern. TP/TC/DP/DC are target/draft prefill/continuation bits. "Y" means full-reference match on that probe in the tested repeats.

Mask	TP	TC	DP	DC	HE15	HE18	HE25	HE31	Panel
0	0	0	0	0	Y	N	N	Y	N
1	1	0	0	0	Y	N	Y	Y	N
2	0	1	0	0	Y	N	Y	Y	N
3	1	1	0	0	Y	Y	Y	Y	Y
4	0	0	1	0	Y	N	N	Y	N
5	1	0	1	0	Y	N	Y	Y	N
6	0	1	1	0	Y	N	Y	Y	N
7	1	1	1	0	Y	Y	Y	Y	Y
8	0	0	0	1	Y	N	N	Y	N
9	1	0	0	1	Y	N	Y	Y	N
10	0	1	0	1	Y	N	Y	Y	N
11	1	1	0	1	Y	Y	Y	Y	Y
12	0	0	1	1	Y	N	N	Y	N
13	1	0	1	1	Y	N	Y	Y	N
14	0	1	1	1	Y	N	Y	Y	N
15	1	1	1	1	Y	Y	Y	Y	Y

Four engineering PQ warm requests failed when graph metadata capacity was insufficient. A resource-only correction scaled metadata reservation by the already declared maximum $q = 4$ without changing numerical operators or the formal panel. SmolLM2/Llama PQ arms use that registered executable; the other arms retain the original route with identical numerical DLLs. Alias-admission and packaging corrections remain in the archive. They are not additional scientific replications.

S2. Complete output and repair-cost results

S2.1. Larger confirmation

The original M_3/C_3 arms use mask3, UB4 and the archived `query_batch` mode label (C_3 also uses $k = 3$). That label does not establish which later PQ/QB graph implementation it represents. The following qualification and cost results retain the historical arm names and are not assigned to QB. The larger campaign has 3,600 complete required requests: three families, five arms, 24 tasks, two adjacent sessions and five repetitions, plus 150 excluded warmups. Its panel was frozen against accessible execution history. M_3 and C_3 each match 240/240 endpoints per family. M_0 differs on 5, 14 and 9 tasks; the Qwen3 failures are GSM8K141,443,562,660,722. The ordinary-production path qualifies for Qwen3 but fails 140 SmolLM2 and 80 Llama requests. Table S2 keeps

Table S2: Larger confirmation ratios. Numerator time / denominator time. Dashes preserve failed qualification.

Comparison	Qwen3	SmolLM2	Llama
R_{prod}/C_3	1.569566	—	—
M_3/C_3	1.134924	0.607389	1.072227
R/C_3	1.569977	0.877174	1.609468
M_0/M_3	—	—	—
R/M_0	—	—	—
R/M_3	1.383332	1.444173	1.501051
R_{prod}/M_3	1.382970	—	—

Table S3: Historical frozen-panel confirmation: matching endpoints out of 240 per arm/family. M_3/C_3 retain their original implementation identity; M_0 is unconstrained UB4 and R_{prod} an engineering-selected ordinary recipe.

Arm	Qwen3	SmolLM2	Llama
R	240	240	240
M_0	190	100	150
M_3	240	240	240
C_3	240	240	240
R_{prod}	240	100	160

failed comparisons as dashes. For M_3/C_3, regressed tasks number 0/24, 24/24 and 4/24. Llama regressions are MBPP418,476,79,99. Failed M_0/M_3 comparisons have diagnostic ratios 0.968456, 0.948603 and 0.936943; failed SmolLM2/Llama R_{prod}/C_3 comparisons are 0.512333 and 0.842790. They remain unqualified because the outputs differ. The 72 separate lite timing observations did not establish mutually exclusive completed GPU stages; their component model remains unavailable.

The ordinary R_{prod} recipe keeps UB1 for Qwen3 and uses UB512 for the other families. Its SmolLM2 and Llama failures prevent those comparisons from being reported as reference-preserving deployment gains. The archived mask3/UB4 implementation is not retrospectively assigned to PQ or QB; the explicitly labelled seven-arm and E2 studies supply that comparison.

S2.2. Seven-arm implementation comparison

The 12-task panel comprises MBPP 123, 140, 255, 401, 413, 418 and GSM8K 111, 120, 1308, 141, 322, 443. Two adjacent sessions and four repetitions per arm/family give 2,016 formal requests and 96 qualified endpoints per arm/family. Table S4 gives all 21 ratios. Figure S1 retains the task-level local-repair and draft comparisons.

Qwen3 target-only local-repair regressions are GSM8K141 and 322. Llama regresses on MBPP418 for target-only and GSM8K120,141 for speculative local repair. SmolLM2 favors QB on all tasks in both local comparisons. Every GS/QB task favors QB. For the draft increment, Qwen3 regresses on MBPP413, Llama on MBPP418, and SmolLM2 on all 12 tasks. SmolLM2's ordinary-to-candidate comparison regresses on 10/12.

The original session-specific ratios are retained below. Target-only Qwen3 and Llama reverse direction; speculative comparisons favor QB in both adjacent sessions. E2 tests these small effects with a different, independently scheduled design and is reported separately. All original task-level and session values are retained in `anc/plot_values.json`. The paired-ratio identities used for draft value assume the same complete panel and weights. They do not estimate attention, projection, feed-forward or synchronization shares of request time.

Table S4: All 21 qualified implementation comparisons at reported six-decimal precision. Regressions count tasks on which the denominator is slower.

Family	Comparison	Ratio	Regressions
Qwen3	$M_{\text{PQ}}/M_{\text{QB}}$	1.026676	2/12
Qwen3	$C_{\text{PQ}}/C_{\text{QB}}$	1.043676	0/12
Qwen3	$M_{\text{GS}}/M_{\text{QB}}$	1.330106	0/12
Qwen3	$C_{\text{GS}}/C_{\text{QB}}$	2.143525	0/12
Qwen3	$M_{\text{QB}}/C_{\text{QB}}$	1.143749	1/12
Qwen3	R/M_{QB}	1.334051	0/12
Qwen3	R/C_{QB}	1.525820	0/12
SmolLM2	$M_{\text{PQ}}/M_{\text{QB}}$	1.038478	0/12
SmolLM2	$C_{\text{PQ}}/C_{\text{QB}}$	1.068111	0/12
SmolLM2	$M_{\text{GS}}/M_{\text{QB}}$	1.366008	0/12
SmolLM2	$C_{\text{GS}}/C_{\text{QB}}$	1.462530	0/12
SmolLM2	$M_{\text{QB}}/C_{\text{QB}}$	0.633948	12/12
SmolLM2	R/M_{QB}	1.369890	0/12
SmolLM2	R/C_{QB}	0.868439	10/12
Llama	$M_{\text{PQ}}/M_{\text{QB}}$	1.013407	1/12
Llama	$C_{\text{PQ}}/C_{\text{QB}}$	1.021848	2/12
Llama	$M_{\text{GS}}/M_{\text{QB}}$	1.411508	0/12
Llama	$C_{\text{GS}}/C_{\text{QB}}$	2.175941	0/12
Llama	$M_{\text{QB}}/C_{\text{QB}}$	1.116620	1/12
Llama	R/M_{QB}	1.426161	0/12
Llama	R/C_{QB}	1.592480	0/12

Table S5: Original two-adjacent-session PQ/QB ratios.

Family	Path	Session 0	Session 1
Qwen3	M	0.971380	1.085118
Qwen3	C	1.041728	1.045626
SmolLM2	M	1.040019	1.036939
SmolLM2	C	1.066179	1.070046
Llama	M	0.994477	1.032698
Llama	C	1.016460	1.027264

Table S6: Retrospective repair–draft contrasts reconstructed from Table S4. $D_L > 1$ favors drafting; Γ compares draft value under QB and GS. These are common-panel ratios, not operator-time fractions.

Family	D_{GS}	D_{PQ}	D_{QB}	Γ
Qwen3	0.710	1.125	1.144	1.612
SmolLM2	0.592	0.616	0.634	1.071
Llama	0.724	1.107	1.117	1.542

Table S6 makes the repair–draft comparison explicit using the existing common-panel estimates. The source values are reported to six decimal places. Propagating their rounding bounds leaves the displayed three-decimal contrasts and all draft-parity classifications unchanged. These numerical rounding bounds are separate from sampling uncertainty; the retrospective contrast adds neither requests

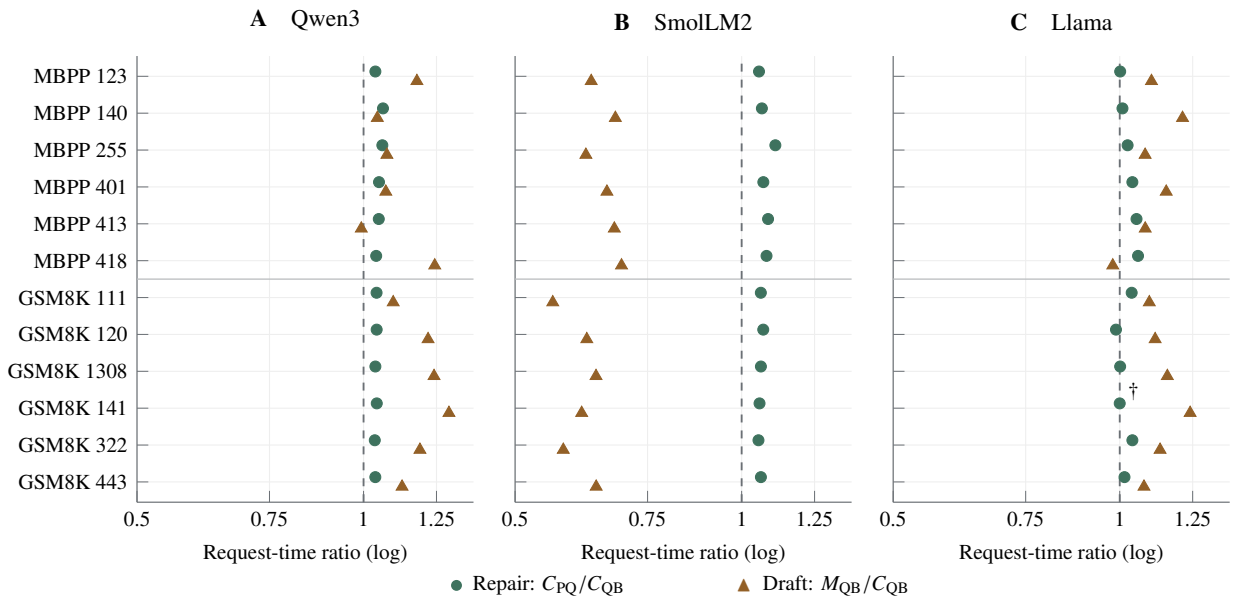


Figure S1: Original repair campaign: task-level local-repair benefit C_{PQ}/C_{QB} (circles) and draft increment M_{QB}/C_{QB} (triangles). Session medians are combined geometrically. The dagger marks Llama GSM8K141, whose local-repair ratio is 0.999919. These are not E2 observations.

nor a new confidence interval.

S2.3. Original-response audit of logical work

The retrospective audit reopens the original response records for all 2,016 formal requests and excludes 42 warmups. Matching family, input, session and repetition gives 288 matched seven-arm audit groups. Each group contributes one GS/PQ/QB triple for target-only execution and one for speculative execution: 576 triples and 1,728 within-path pairwise comparisons. The exported logical-work counters and acceptance/caller-width histograms agree in every comparison (Table S7). Complete native token-ID arrays, recorded EOG flags, length, terminal ID, stop reason, cache count and truncation flag also agree in all 1,728 non-reference-to-R endpoint comparisons. Each original record's normalized IDs and workload are checked against its native response fields.

The original execution has 42 family/session/arm workers, each with 48 formal requests; the second session reverses the arm order. The 288 audit groups are reconstructed by task and repetition across these workers, not collected as adjacent randomized seven-arm time blocks. Matching their recorded work does not remove between-worker temporal drift.

The counter definitions distinguish caller work from physical execution. Both GS/local contrasts change physical graph calls, graph reuse and target prefill widths in all 288 pairs per path; speculative execution also changes continuation physical-width histograms. The draft's physical counters remain equal. PQ/QB agree in every exported workload field. These are aggregate counts and histograms, not complete per-cycle trace or bitwise state comparisons. They establish matched recorded work without assigning the request-time difference to projection, attention, feed-forward computation or instrumentation.

The source manifest pins all archive parts. This audit retrieves and checks parts 001, 007 and 009 by their committed Git-blob and SHA256 identities, then verifies each selected original member against the archived index. It does not recompute the whole nine-part ZIP hash. The paper includes a compressed field projection with complete endpoint arrays, original record hashes, the frozen plan, counter definitions and the original index under `anc/seven_arm_work_audit/`. The reconstruction script recomputes all pairings and conservation identities from that projection. Its original-record and split-archive modes additionally reopen the archived bytes and reproduce the projection exactly. This adds no sessions and leaves all original timings and statistical weights unchanged.

Freeze versions and executable identities are distinct. The prospective v02 correction retained the 96 completed Qwen3 session-0 R/M_{QB} records under their initial freeze; the remaining 1,920 records carry v02. Actual worker commands use the metadata-capacity-corrected executable only for SmolLM2 and Llama PQ arms (384 formal requests), while the other 1,632 use the original executable. The correction enlarges PQ metadata reservation; all 21 numerical DLLs and the counter/observer sources retain identical hashes. GS/QB therefore share an executable, while the PQ comparison includes its recorded metadata-capacity correction. Exact routes and source differences remain in the audit receipt.

S3. RTX replication and numerical replay

S3.1. E2 design and complete estimates

The replication retains the 12-task implementation panel and the original reference, models, sampler, timer and runtime routes. Each session has 54 fresh workers: short PQ/QB blocks and serial-reference blocks, with one registered warmup per worker. Across six sessions this gives 864 PQ/QB requests, 432 references and 324 warmups. Each task/session has two matching serial-reference endpoints. All required endpoints and observation checks pass; no latency is trimmed.

Family	Matched groups	M work triples	C work triples	Endpoints vs. R
Qwen3	96	96/96	96/96	576/576
SmolLM2	96	96/96	96/96	576/576
Llama	96	96/96	96/96	576/576
Total	288	288/288	288/288	1728/1728

Table S7: Retrospective audit of all 2,016 formal original seven-arm requests, excluding 42 warmups. Each audit group matches family, task, session and repeat across arm-specific workers; each work triple compares GS, PQ and QB within the target-only (M) or speculative (C) path. All three pairings per triple agree in the recorded logical-work counters and acceptance/width histograms (1,728 pairings total). Endpoint checks compare the complete native token IDs and recorded EOG flags, stop reason and terminal metadata with the paired R request. Physical graph calls, widths and reuse are reported separately; aggregate counter equality does not establish bitwise state or per-cycle trace equality. This adds no sessions and changes no original timings or weights.

The fixed runtime routes use the original executable for every Qwen3 arm and all R/QB arms. SmolLM2 and Llama PQ arms use the recorded metadata4 correction; both builds retain identical hashes for all 21 numerical DLLs. The source/build manifests identify each route. The launcher clears inherited GGML/CUDA environment overrides, but neither that configuration nor GGML topology reuse establishes actual CUDA Graph replay. Per-request capture, replay and fallback counts are absent from the recovered E2 report records.

`E2_REQUEST_LATENCIES.csv` retains all 1,620 rows, including the 324 labelled warmups, with its original Git-blob identity. `generate_e2_absolute_cost.py` checks the source and reconstructs Table 4. The per-arm geometric means use the same 12-task and six-session weights as Equation 7, so their quotients reproduce the paired ratios. The table's `latency_ms` is the resident request interval; raw start/end nanosecond fields remain in the full request archive.

All consecutive session gaps exceed one hour; first-to-last start is 24.450801 hours. The original interpreter bytes were restored after a pre-start identity mismatch before the final session. The original freeze checks then passed without exception and the untouched scientific slots ran. The earlier stopped record remains part of the history.

Table S8 gives the complete six-session estimates. The intervals use the original random seed 20261014, 10,000 bootstrap draws of six session log means, and linearly interpolated 2.5/97.5 percentiles. They quantify fixed-panel, single-host variation, not a date-level population or multiple-comparison guarantee. Table S9 gives all 72 task aggregates; 9 regress. Full session contrasts and observed session times are in `anc/rtx104/`.

Recorded acceptance, rollback and logical-workload counters agree in all 216 speculative task/session pairs. This comparison is limited to the recorded counters and histograms, rather than complete control-trace identity. Model callers and physical micrographs remain distinct from hardware launches.

S3.2. Sensitivity to interval method and session omission

The registered percentile bootstrap remains the primary analysis. Table S10 adds a post-review t interval,

$$\exp\left(\bar{z} \pm t_{0.975,5}\frac{s_z}{\sqrt{6}}\right), \qquad t_{0.975,5} = 2.570581835636314,$$

where z_s is the session mean log contrast and s_z its sample standard deviation. This analysis assumes independent, approximately normal session log means; it does not add inputs, dates or hosts. Six further point estimates omit one session each.

All speculative t intervals remain above parity. Qwen3 and SmolLM2 also remain above the practical ratio threshold 1.02; Llama crosses it. Its omit-one estimates all exceed 1.02. Thus the direction of the Llama effect is stable across these checks, while its practical-threshold conclusion depends on the interval method. The source CSV is unchanged; `scripts/generate_e2_sensitivity.py` reconstructs the new table and the complete omit-one estimates.

S3.3. Retrospective draft-on/off analysis

For each family, session and input, the reanalysis pairs exactly one non-warmup record from each of $M_{PQ}, M_{QB}, C_{PQ}, C_{QB}$. There are 216 complete four-arm cells (864 measured requests). The 324 registered warmups and 432 serial-reference measurements are excluded from these contrasts after their source accounting and recorded endpoint metadata are checked. Each session averages twelve task log ratios; the reported estimate averages the six session means. Pointwise exploratory t_5 intervals use the six session log means with no multiplicity adjustment. These contrasts were selected retrospectively and do not replace the original primary E2 comparisons or their registered bootstrap analysis. These intervals retain the fixed-input, single-host scope of E2 and do not add a native execution or repeat full token-array qualification. The saved block order is M before C in four of six Qwen3 sessions, three of six SmolLM2 sessions and four of six Llama sessions; the remaining sessions reverse that order. Thus the M/C contrasts use separately scheduled within-session blocks, not newly randomized adjacent per-input M/C measurements, and do not isolate block-order or temporal-drift effects.

The local-layout interaction is $\Gamma = D_{QB}/D_{PQ} = (C_{PQ}/C_{QB})/(M_{PQ}/M_{QB})$. Its estimates are 1.010828, 1.037008 and 1.013313 for Qwen3, SmolLM2 and Llama. The exploratory intervals span parity for Qwen3 and Llama; SmolLM2's interval is [1.027962,1.046134]. Thus an improvement in draft value from QB does not itself imply profitable drafting, as the SmolLM2 result demonstrates.

The six-session QB task aggregates favor drafting for 10/12 Qwen3 inputs, 11/12 Llama inputs and 0/12 SmolLM2 inputs. These task contrasts remain descriptive. The source-pinned reconstruction script

Table S8: All completed E2 comparisons. Ratios favor QB above one; regressions count six-session task point estimates below one.

Family	Path	Ratio	Lower 95%	Upper 95%	Regressions	Practical threshold 1.02
Qwen3	M	1.020563	1.013023	1.026553	2/12	Point above; interval overlaps
Qwen3	C	1.031614	1.025393	1.037926	1/12	Interval above
SmolLM2	M	1.027767	1.021010	1.035017	2/12	Interval above
SmolLM2	C	1.065803	1.063387	1.068399	0/12	Interval above
Llama	M	1.015593	1.009627	1.021622	3/12	Point below; interval overlaps
Llama	C	1.029114	1.020126	1.039149	1/12	Interval above

Table S9: E2 task-level PQ/QB ratios, geometrically averaged over six sessions. Values below one are italicized; these are descriptive task contrasts, not significance tests.

	Qwen3		SmolLM2		Llama	
Task	M	C	M	C	M	C
GSM8K 111	1.015305	1.028626	1.031259	1.064372	1.027335	1.029118
GSM8K 120	1.007356	1.073052	1.050246	1.073224	1.021427	1.026504
GSM8K 1308	1.018527	1.030064	1.057675	1.070511	1.020762	1.041848
GSM8K 141	*0.992667*	1.035252	1.021013	1.051986	1.012332	1.028335
GSM8K 322	1.001169	1.033028	1.028420	1.059544	1.026791	*0.993618*
GSM8K 443	1.003752	1.021251	1.033367	1.067165	1.024666	1.019558
MBPP 123	1.056908	1.037495	1.044148	1.058628	*0.992752*	1.023365
MBPP 140	*0.981806*	1.052058	1.041709	1.063535	*0.978786*	1.048076
MBPP 255	1.079294	1.032638	*0.972948*	1.090371	1.021555	1.054405
MBPP 401	1.001954	1.024979	1.044272	1.048708	1.026712	1.009394
MBPP 413	1.089971	*0.967860*	*0.978119*	1.089698	*0.986121*	1.035206
MBPP 418	1.004440	1.046335	1.033875	1.052829	1.050112	1.041465

Table S10: E2 session sensitivity on the fixed 12-input panel. Ratios are PQ/QB; M is target-only and C is speculative. The registered bootstrap remains primary. Secondary t_5 intervals assume independent, approximately normal session log means. The final column gives the range of six leave-one-session-out point estimates.

Family	Path	Estimate	Primary bootstrap 95% CI	Secondary t_5 95% CI	Omit-one range
Qwen3	M	1.020563	[1.013023, 1.026553]	[1.011013, 1.030204]	[1.018397, 1.023944]
Qwen3	C	1.031614	[1.025393, 1.037926]	[1.022587, 1.040720]	[1.029113, 1.034273]
SmolLM2	M	1.027767	[1.021010, 1.035017]	[1.017648, 1.037987]	[1.024785, 1.030395]
SmolLM2	C	1.065803	[1.063387, 1.068399]	[1.062196, 1.069422]	[1.064882, 1.066443]
Llama	M	1.015593	[1.009627, 1.021622]	[1.006830, 1.024433]	[1.013457, 1.018003]
Llama	C	1.029114	[1.020126, 1.039149]	[1.015387, 1.043026]	[1.024960, 1.032333]

`generate_e2_draft_profitability.py` verifies the complete quartet structure and generates all estimates, session rows and source-line pairings.

S3.4. E1 coverage and controls

The independently frozen diagnostic retains all 21 numerical DLLs and changes only the two admitted capture/replay source files. Its 72 contexts use target layers $0/\lfloor L/2 \rfloor/(L-1)$, both phases and $q = 1$–4. One exposed text, GSM8K443, is selected by longest inherited prompt with lexical tie-breaking for each tokenizer. It has 105/126/123 prompt IDs for Qwen3/SmolLM2/Llama. Each context has a fresh worker with a counts warmup, heavy capture and counts control; each observed MQ/PQ/QB graph is replayed twice. All 72 contexts are observed and admitted; all requests and 432 operator starts complete.

The physical key extent is 256 in every capture. The per-query visible mask counts are 1 through q in prefill, 106 through $105 + q$ for Qwen3 continuation, 127 through $126 + q$ for SmolLM2, and 124 through $123 + q$ for Llama. Minimum-key selectors refer to tensor extent, not unmasked context; this is not a query-width by visible-length factorial. All exact coordinates are in `E1_CONTEXTS.csv`.

PQ/QB bitwise equality is 72/72 at each stage. MQ/PQ and MQ/QB equality is 18/72, exclusively the $q = 1$ controls. Table S12 retains the largest measured differences. These compare operator arrays, not next-token logits.

Large ULP distances near zero or across signs are retained rather than converted into a tolerance-based pass. Input identity is checked before and after replay against frozen hashes. Native worker module receipts are retained; the inherited standalone replay binds executable and DLL paths but does not add a separate loaded-module inventory. Kernel symbols, reduction trees and continuous residency

Table S11: Exploratory E2 draft and layout–draft contrasts. Pointwise 95% t_5 intervals use six session mean log ratios and assume independent, approximately normal session means; they are not multiplicity-adjusted. $\Gamma = D_{QB}/D_{PQ}$ measures the change in draft value between the two local layouts, not GS/local repair interaction.

Family	Contrast	Estimate	Pointwise 95% interval	Sessions > 1	Sessions < 1
Qwen3	D_{PQ}	1.072058	[1.058882, 1.085397]	6/6	0/6
Qwen3	D_{QB}	1.083666	[1.059328, 1.108563]	6/6	0/6
Qwen3	Γ	1.010828	[0.996111, 1.025762]	4/6	2/6
SmolLM2	D_{PQ}	0.575629	[0.563961, 0.587538]	0/6	6/6
SmolLM2	D_{QB}	0.596931	[0.582288, 0.611943]	0/6	6/6
SmolLM2	Γ	1.037008	[1.027962, 1.046134]	6/6	0/6
Llama	D_{PQ}	1.090530	[1.069862, 1.111598]	6/6	0/6
Llama	D_{QB}	1.105048	[1.096790, 1.113369]	6/6	0/6
Llama	Γ	1.013313	[0.994814, 1.032156]	5/6	1/6

Table S12: E1 maximum MQ-to-repair differences; both repairs give the same values. PQ/QB has zero error and ULP distance at all stages.

Stage	Max. finite absolute	Max. finite ULP
QK	0.078582763671875	2,017,357,824
Softmax	0.0009172558784484863	96,633
AV	0.010433197021484375	1,953,885,696

remain unidentified.

S3.5. Restoration and interpretation

The host downloaded and restored the authenticated E2, E1 and control archives into a fresh directory, verifying 10,179, 9,696 and 47 members. A separate interpreter reconstructed 11 analysis outputs byte-for-byte. This is same-host restoration, not an external replication. This manuscript's own checker reconstructs the six ratios and bootstrap intervals from the submitted session table, verifies the copied context/time-table identities and checks the report summaries; it does not replay the original tensor archives.

S4. Vulkan partition and retained-state experiments

S4.1. Inputs and partition controls

A uses UB512/capacity one and a whole-prompt submission; B retains that context and externally submits 128 + remainder; C uses UB128 with internal partitioning. Only the final absolute prompt position requests logits. Three repeats per input/arm, one capture-off and two capture-on, give 63 requests. All B/C initial full FP32 logits, source-defined sequence-KV payloads and complete endpoints agree. Initial state comprises 64 F16 K/V tensors over 32 layers and recorded logical positions. Full KV payloads are retained at prefill completion; later cuts contain full logits but only KV hashes. The synthetic prefix129 difference at 57 was not added retrospectively as a capture cut.

The final follow-up contains 21 fourth-arm observations, 38 partition-sweep observations, 18 native captures, 16 shams and 40 crosses: 133 whole requests, with a zero-forward load probe separately counted. All 19 schedules are listed in Table S14. Driver-body hashes were authenticated only after the fourth-arm and sweep stages, which remain diagnostic rather than fully compliant confirmation. Authentication preceded the operator and restoration stages.

Table S13: Seven original partition inputs. All B/C full endpoints and initial FP32/KV payloads agree. The first difference is A/B, zero-based.

Input	Prompt IDs	First difference	A/B initial state
GSM1248	161	47	Different
GSM831	133	81	Different
GSM1011	190	—	Different
GSM1249	163	—	Different
Prefix 1248/127	127	—	Equal
Prefix 1248/128	128	—	Equal
Prefix 1248/129	129	57	Different

Table S14: All frozen exposed partition schedules. The final column is first output difference from A; no numerical monotonicity is inferred.

Input	Caller widths	Endpoint class	First diff.
1248	64+64+33	A	—
1248	96+65	A	—
1248	128+33	B/C	47
1248	160+1	A	—
831	64+64+5	A	—
831	96+37	A	—
831	128+5	B/C	81
831	132+1	Third class	40
1011	64+64+62	All	—
1011	96+94	All	—
1011	128+62	All	—
1011	160+30	All	—
1249	64+64+35	All	—
1249	96+67	All	—
1249	128+35	All	—
1249	160+3	All	—
Prefix 129	64+65	A	—
Prefix 129	96+33	A	—
Prefix 129	128+1	B/C	57

S4.2. Projection and boundary reconstruction

Native comparisons align actual embedding inputs, logical positions and the intended pre-attention RMS producer.

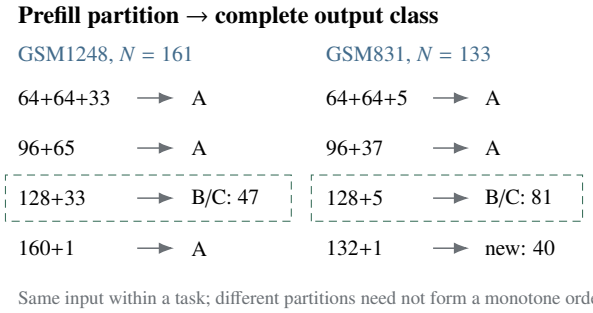


Figure S2: Selected partition classes from the complete schedule table. A is whole prefill; B/C is the 128-split class. Numbers are first output differences from A. The map does not imply a universal 128-token threshold.

Table S15: Native layer-0 A/B projection differences. Values are maximum absolute differences for the stated tensors, not model-logit perturbation bounds.

Projection	1248	831
Q	0	0.0207662582
K	0.0527343750	0.0703125000
V	0.0023193359	0.0021639541

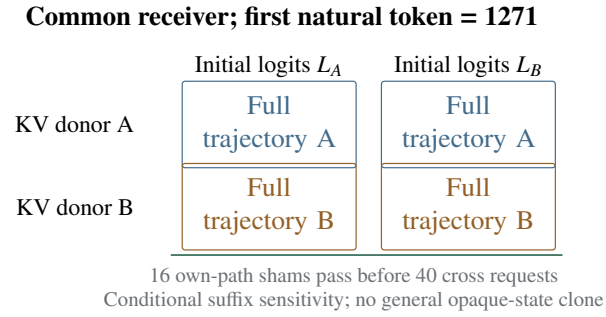


Figure S3: Admitted boundary0 cross for the affected tasks. All initial logits select 1271. The subsequent suffix follows the KV source under the same reconstructed receiver; stable controls retain their endpoint.

Table S15 reports the layer-0 projection differences. Listing sibling projections in an audit sequence does not identify a unique first-cause kernel. Forty standalone calls compare RMS, scale/MUL and Q/K/V projections on two full/split operand geometries, with two repeats. Thirty-two pass native same-arm correspondence; eight scale/MUL calls fail because the captured input aliases a buffer already changed by an in-place consumer. Those failures remain excluded. RMS and the first task's Q controls agree bitwise, while admitted differing projection replays reproduce geometry sensitivity.

The common receiver restores sequence KV and positions, applies the native sampler to a complete 128,256-element initial FP32 vector, and reconstructs the empty generated prefix and pending token. Dynamic output buffers are rebuilt; opaque graph-cache pointers are not cloned. Sixteen own-path shams cover four inputs, both donor states and two repeats. Forty crosses then use three repeats per cell on the two affected tasks and two on the stable controls.

S4.3. Separate physical-work and cost boundary

An earlier speculative actor executes a logical seed-plus-proposal verification as two occupied width-one target micrographs at $k = 1$. The work-witness study retains 563 successful cycles and separate draft counts; one API-observer request failed a logging threshold and was not repeated. Its partial output is not a GPU timeline.

The original stock-relative cost campaign failed reference qualification. A later post-hoc full-output audit supports a different production-path pairing on 24 tasks and 240 repetitions: $R_{\text{prod}}/C = 0.719264$ and $M_{\text{match}}/C = 0.727927$, with every task regressing. Those negative same-output economics do not repair the original stock gate. They are not remeasured by the causal studies, and no completed component-time model is admitted.

S5. State components and decision frontiers

S5.1. Selectors and complete endpoint matrix

A denotes the historical whole-prefill state and B the partitioned state. Each intervention preserves recipient initial logits, natural pending-token consumption and the common receiver. Thirty-two EMPTY/ALL controls pass before the 160 subset requests. Table S16 retains all 80 task/selector/direction coordinates; both process repeats agree. When donor and recipient natural endpoints coincide, a match is D rather than attributed to a distinguishable donor. All 76 D requests change actual bytes. K-only or V-only layer0 selects 1/64 of K/V value storage, excluding metadata. Early, middle and late groups contain 8, 16 and 8 layers. Prefix is $[0, 128)$; tail is its complement, with five positions for /831. K is token-major and non-flash V feature-major, so each selector uses the actual storage strides. The source records separately retain selected bytes, changed bytes and numerical distances.

S5.2. Magnitude and common-prefix selection

Table S17 reports differences computed from stored F16 values using the actual A-state norm. No increasing one-threshold rule perfectly separates these four endpoint labels: a smaller overall perturbation can accompany divergence while a larger one does not. This is a finite counterexample to a magnitude-only explanation, not an alternative predictor. For each task, recipient and panel, the frontier t^* is the first output at which the four natural trajectories cease to agree. Their consumed prefix is identical at that point. L0 and POSITION have separate frontiers; /831 recipient A reaches them at 40 and 81 respectively. Later vectors cannot fill an earlier missing comparison.

Archived captures support 44 of 112 planned repeat-level contrasts, with 22 agreeing pairs. The 68 incomplete comparisons contain 272 missing panel entries or 240 distinct arm/repeat/position vectors, because some arms are shared. A vector hash or top-two scores do not replace full logits. Output zero uses the frozen initial logits and needs no additional forward. Table S18 retains every available frontier contrast. For /1248 B→A L0 at 47, the additive joint prediction is 0.08137702941894531, the actual joint is −0.05448722839355469 and $I_{KV} = -0.1358642578125$. Full-vocabulary argmax is 220/220/832/220. At output46,

Table S16: All 80 distinct subset/direction coordinates. Both process repeats agree in every cell. Within each task the columns are A→B and B→A. B, C and D denote endpoint classes, not task-answer quality.

	/1248		/831		/1011		/1249	
Selector	A→B	B→A	A→B	B→A	A→B	B→A	A→B	B→A
K_0	B	D	C	C	C	C	D	D
V_0	B	B	B	C	C	D	D	D
$K_0 + V_0$	B	D	B	C	C	D	D	D
All K	D	B	C	C	C	C	D	D
All V	D	B	C	C	C	C	D	D
Early KV	D	D	B	C	C	D	D	D
Middle KV	D	D	B	D	D	C	D	D
Late KV	C	B	D	C	C	C	D	D
Prefix KV	D	B	B	D	C	C	D	D
Tail KV	D	B	B	D	C	C	D	D

Table S17: Full-KV differences, B relative to A, computed from stored F16 values. Endpoint labels refer to natural A/B continuations on four exposed inputs.

Task	Relative L_2	Max. absolute	A/B endpoint
/1248	0.006488689	0.255859375	Different
/1249	0.006748138	0.246093750	Same
/1011	0.006887112	0.265625000	Same
/831	0.010452955	1.332580566	Different

the same fixed-pair residual is −0.13482666015625, but the actual top two are 11/220 with margin 8.017614364624023; the frontier pair 220/832 is not that position's top two. Cross-position comparison is descriptive.

At output1 of /1011, EMPTY margins are 0.561777115/0.549312592, maximum observed $\epsilon_\infty = 0.207827211$, and every arm satisfies $m > 2\epsilon_\infty$ and selects 11886. The mixed trajectories later diverge at 10, whose adjacent full vectors are missing. For /1249 the corresponding margins are 0.959617615/0.936807632, maximum $\epsilon_\infty = 0.194167793$, and every observed arm selects 1505. This sufficient top-1 condition applies only at the measured position.

S5.3. Direction symmetry and independent conditions

For complete states (A_P, A_T) and (B_P, B_T), reversing donor and recipient permutes the POSITION quartet from AA, BA, AB, BB to BB, AB, BA, AA. When the actual state, prefix and receiver agree under this mapping,

$$\boldsymbol{I} = z_{\mathrm{ALL}} - z_P - z_T + z_{\mathrm{EMPTY}} \tag{S1}$$

is invariant. The scalar $I(u) - I(r)$ changes sign when (r, u) is reversed. The audit checks actual state, FP32 vectors, prefixes, sampling conditions and F64 bytes in all 12 available direction pairs. For /1248 at47 both directions give −0.09655952453613281 in pair order (220, 832), despite opposite original signs.

Of the 44 oriented repeat rows, 24 are POSITION and 20 L0. Accounting for direction symmetry and two repeats gives six POSITION and ten oriented L0 conditions, still only four underlying inputs. L0 has different surrounding recipient states and is not collapsed by the POSITION identity.

S5.4. Acquisition failure and data scope

A new acquisition attempted one request with a registry key missing recipient identity. The planned EMPTY B→A imported B rather than A, completing 127 decodes before failing the input check. It is neither a qualified EMPTY nor an ALL sample; 167 slots were unstarted. The later CPU revision repairs request assembly without new native data. The 54 unique FP32 vectors and 16 residual vectors support the historical 44 contrasts; no missing frontier or independent-machine GPU replication is supplied by that revision.

S6. Natural-execution port and backend audit

S6.1. Design and complete exploratory endpoints

The natural pilot keeps the original ordinary UB128/capacity-one reference R. R_SHADOW tests the rebuilt actor without replacing R. Candidate prefill uses reference128 or bulk512 organization, MQ attention and capacity4. Six speculative arms use continuation MQ/PQ/QB, target UB4, $k = 1$ and the same draft; two arms are draft-free. Every request starts from its prompt without imported state.

The planned 80 requests are eight original R, four shadow counts, four shadow diagnostics, 32 candidate counts and 32 candidate diagnostics. It stops with 49 attempts, 48 complete, one interrupted and 31 unstarted. Table S19 preserves all 14 endpoint counterexamples. The reference and shadow outputs agree, but candidate observer pairs are incomplete. The q1 branch bypasses the multi-query transformation, so draft-free QB-labelled agreement does not qualify lifted QB. The frozen monitoring declaration requires 100ms request-period VRAM sampling. The original R instead supplies four checkpoints, with maximum gaps 2874–3033ms. The mismatch is detected while the 49th request is active; only its owned worker is terminated. No sampled threshold violation was observed, but the original declaration remains unmet. P2/P3 are not executed and qualified speedup remains null. The 8,110 caller events, full times and raw accounting are retained in the ancillary record.

The available candidate diagnostics contain histograms but no ordered, phase-bound multirow VERIFY witness,

Table S18: All available frontier contrasts. L0 components are K/V; POSITION components are prefix/tail. Both historical repeats agree. Values are rounded to six decimals. Token pairs are fixed within each panel; direction-dependent coordinates must be canonicalized before comparing signs.

Task	Direction	Panel	t^*	Δ_1	Δ_2	I	EMPTY margin
1248	B→A	L0	47	.059744	.088148	-.135864	.066515
1248	A→B	L0	47	.025909	.033445	-.019032	.010706
1248	B→A	POSITION	47	.078003	.095778	-.096560	.066515
1248	A→B	POSITION	47	-.018557	-.000782	.096560	.010706
831	B→A	POSITION	81	-.095177	-.062551	.238718	.013084
831	A→B	L0	81	.118263	.172575	-.049446	.067905
831	A→B	POSITION	81	.143541	.176167	-.238718	.067905

Table S19: Complete candidate counts endpoints observed before stopping. Values are first differing output indices from original R; "=" denotes an equal complete cap128 endpoint. Candidate observer pairs are incomplete. The execution-contract limitation prevents qualification of this panel.

Arm	/1248	/831	/1011	/1249	Matches
$M_{B,\mathrm{QB}}$	47	81	=	=	2/4
$M_{R,\mathrm{QB}}$	=	=	=	=	4/4
$C_{B,\mathrm{MQ}}$	47	40	10	=	1/4
$C_{B,\mathrm{PQ}}$	47	81	=	=	2/4
$C_{B,\mathrm{QB}}$	47	40	10	=	1/4
$C_{R,\mathrm{MQ}}$	=	40	10	=	2/4
$C_{R,\mathrm{PQ}}$	=	=	=	=	4/4
$C_{R,\mathrm{QB}}$	=	81	10	=	2/4

leaving actual multirow placement unresolved. Descriptive request times include the new actor's observation, lifecycle and phase-management costs as well as attention.

S6.2. Independent representation audit

The follow-up uses the same frozen CPU numerical libraries across 144 development and 288 locked held-out synthetic fixtures. It covers $q \in \{1, 2\}$, key extents 127/128/129/255/256/257, head pairs $(8, 8), (8, 1), (32, 8)$, $d = 128$, and compact or legally offset model-permuted Q. Three nonempty masks and two fixed data patterns give 72 geometries and 432 fixtures; MQ/PQ/QB require 1,296 CPU graph evaluations. Held-out here refers to synthetic test conditions, not new model prompts.

An independent integer oracle checks addresses, contiguous-group GQA, mask columns and output recovery. Four injected faults are detected. A separate FP64 scalar oracle uses stored inputs. Structured one-hot and single-valid-key tests have exact expectations. No strict assertion, crash or nonfinite failure occurs. General differences remain descriptive: maximum absolute error is zero at QK on the dyadic inputs, 5.3608031×10^{-9} at softmax, and 5.5132505×10^{-5} at AV/final output. No post-hoc tolerance defines a general numerical pass. F16-representable dyadic data do not establish generic F32 reduction invariance.

The complete historical endpoints satisfy $M_{B,\mathrm{QB}} = C_{B,\mathrm{PQ}}$, $M_{R,\mathrm{QB}} = C_{R,\mathrm{PQ}}$ and $C_{B,\mathrm{MQ}} = C_{B,\mathrm{QB}}$ on all four windows. On /1011, $C_{R,\mathrm{MQ}}$ and $C_{R,\mathrm{QB}}$ both first differ at10 but have different full endpoints. These relations do not establish common operands, KV or acceptance traces.

S6.3. Version-bound support semantics

Seven core backend/header hashes match the archived build manifest and the numerical DLLs are unchanged. The frozen `ggml_backend_vk_device_supports_op` rejects the reviewed QK and AV shapes because their fourth source extents are $(1, 2)$. The complete source identity and line location are in `anc/portability111.json`. Unbound shader bodies are excluded from arithmetic attribution.

The descriptor checker rejects the unequal fourth extents and leaves other cases unresolved because equality is only one necessary admission condition. This CPU/source analysis identifies the backend change required before a same-backend numerical comparison. Actual routing and fallback remain unknown, and the stopped pilot retains its failed qualification.

S7. Mapping, representation and observer controls

The post-replication mapping studies retain distinct references, input sets, runtimes and timers [20, 21]. Their results are not additional samples of the original E2 experiment. The full scientific heads and paths are in `anc/source_manifest.json` and the follow-up evidence index.

S7.1. Input lifetime and software controls

The original RTX allocation allowed later AV and ordinary-output nodes to overwrite storage holding K/V. Address and byte witnesses, re-upload interventions and return to the original allocation localized this failure. Independent persistent input storage restored repeated input and output bytes. This repair establishes valid repeated execution; it supplies no accelerator result.

Balanced PQ/QB/QH comparisons use identical QB_A/QB_B controls. The core $q = 2, 4$ panel yields QB_A/QH=0.900826, with 19 of 24 geometries supporting lower QB cost under the declared descriptive screen and five unresolved. A/A pilot envelope reduction to a median 2.762% does not establish 2% precision for each geometry. A selector predicts all 12 held-out synthetic mean directions but costs 1.010298 times fixed QB once the measured lookup is charged; its estimate is an offline phase sum, not an integrated request.

CUDA Graph ON is established through capture, instantiation and replay in instrumented hosts. Clean hosts share the graph/runtime identity without loading the observer.

The ON/OFF intervention reduces QB_A completed-graph cost to 0.615–0.849 of OFF. Profiling traces identify path changes, but kernel durations do not supply a union of busy time or a request critical path.

S7.2. Fixed-logical-input padding and cost windows

Logical-length-preserving padding keeps original mask columns and appends finite zero K/V with fully masked keys. L1023/S1024 fails all four reference qualifications despite reaching three kernels at the first observed execution. L1023/S1025 qualifies all four, retains eight kernels and costs 1.014783/1.014974 times the two native controls. L1024/S1025 and L1024/S1026 also fail the exact reference. All failed paths stop at the first mismatch; no clean steady-state benefit is reported for them. Source-to-DLL correspondence for the internal RTX divergence remains unknown.

The complete RTX Q transaction begins with common permuted host Q and includes copy/transpose, Q upload/completion, graph replay/synchronization and full ordinary readback. For $q = 2, 4$, compact-QH/QB_A is 1.002216/1.001598 and compact-QH/QB_B is 1.008295/1.004167. K/V/mask are resident and targets preallocated; the interval excludes natural Q generation and continuing KV updates.

The B570 equal-head panel instead uses Hq/Hkv=8/8, $d = 128$, key1023/1024/1025 and $q = 1, 2, 4$. Its narrow admission repair enables the existing batch-broadcast shader without K/V REPEAT. Three fixed input sets and three process sessions, balanced over both identical QB controls and QH, give a resident-graph difference of $-1.028\,\mu$s $[-2.177, +0.067]$. The separately measured complete representation transaction gives $-1.518\,\mu$s $[-4.168, +0.916]$, ratio 0.992244 [0.979908,1.004691]. These are descriptive adjusted intervals from three process clusters on one device/date; all practical-effect screens remain unresolved. The 43,740 transactions are technical repetitions, not independent workloads. The two windows cannot be subtracted to isolate transfer cost.

S7.3. Request relevance and observer integrity

The first B570 Qwen3 request bridge is a separate q1/key256/GQA32:8 observation. It does not exercise the equal-head near1024 micrograph. Serial per-operation logging adds about 12% median decode overhead and fails its 10% observer screen. Concurrent logging initially omits the terminal interval and includes inactive-node labels; terminal timestamps and the executor's active-node predicate repair those faults.

The final same-DLL clean/profile/clean medians are 27.6675/29.6058/27.6794 ms, so the observer still adds about 7%. Each final context table counts 16 output projections and 5,004 QK, softmax, AV and CONT operations each. The third-context GPU subtotal is 3.490070 s over prefill and continuation. Pure QK/softmax/AV contributes 2.8005%, or at most 2.8026% when the mixed group is included; the core excludes projections, RoPE and KV updates. Quantized matvec/fused groups contribute 92.5152% and contain both attention projections and FFN. These diagnostic fractions are not clean request shares or a bandwidth diagnosis.

S8. Historical finite-panel service results and prefill closeout

S8.1. Panel identities and qualification

Table S20 lists the two service contracts. Each panel has four exposed inputs, $k = 1$, 32/8 target GQA, F16 KV and actual two-query verification in all 36 target layers. The draft and target models are Qwen3-0.6B and Qwen3-8B Q8_0. Model/tokenizer hashes and exact driver identities are pinned separately; the shared family names do not make these a paired GPU experiment.

The RTX source/driver registration precedes every model start; its independently generated R/PQ reference is sealed before qualification, and the qualification record is sealed before timing [16]. R uses its matching reference; PQ and QB use the matching PQ target/draft vectors, including rejected verification rows. Their complete speculative traces also agree. This does not assert that every internal PQ value equals a serial R value. Query geometry and mask extents/strides are observed, whereas visible-key counts are derived from the registered causal positions. No operator readback is inserted into the natural CUDA graph.

The B570 AV repair first passes six synthetic key256/512 by $q = 1, 2, 4$ coordinates with unchanged PQ stage files, then a model integration passes one exposed 16-token bridge. Its first cost pilot preserves a 5.123483-second discrepancy between identical PQ arms. The apparently favorable aggregate is not promoted into a speedup. A separately registered four-input panel uses a 64-token cap and natural EOG, and qualifies complete R/PQ/QB endpoints, 346 PQ/QB vectors and full speculative cycle traces before timing [19, 15]. The old Llama Q6_K references and their failures remain separate.

S8.2. Every input and paired block

The tables and Figure S4 are generated from the saved block records. RTX blocks retain all four arm service times, permitting direct recomputation of each ratio. B570 aggregates are reconstructed from archived paired ratios; the original service logs and full raw logits are outside the manuscript package. `anc/check_closure.py` verifies exact Git blob identities, weights, input/session coverage, ratios, plot coordinates and printed cells.

The primary layout estimator gives equal weight to each input and session in log space. Its A/A criterion is the largest absolute per-input geometric-mean PQ_A/PQ_B deviation from one, not the largest individual block deviation. The finite engineering screen also requires more than 2% reduction and all session aggregates below one. RTX session ratios are 0.979094, 1.038635, 1.005512 and 0.936718, and the A/A criterion is 5.208%. B570 session ratios are 0.996243, 0.995313, 0.996120 and 0.992904, with A/A criterion 0.754%; its 0.486% layout point reduction is below 2%. Neither panel passes the practical layout screen.

All 16 RTX QB/R blocks are below one and all 16 B570 QB/R blocks exceed one. Aggregate ratios are 0.582325 and 2.619763 respectively. RTX dual-PQ/R is 0.588633, showing that the speculative-policy benefit is already present with PQ. The RTX estimate corresponds

Table S20: Distinct service contracts. The entries describe the recorded experiments, not recommended production configurations.

Property	RTX service study	B570 formal request study
Reference and inputs	New R/PQ full-logit references, historical native endpoints; MBPP/413, GSM8K/120,322,443	Prospective q1 reference; HumanEval/38,65,115,127 with their recorded chat formatting
Output window	Native EOG or cap128	Native EOG or cap64
Target/draft capacity	R target UB1; speculative target UB4; draft UB1	Target verification UB4; draft UB1; serialized q1 prefill
Context and host policy	Context1024, 24 host threads	Context1024, eight context thread parameters
Measurement	64 requests in four timing processes; 64 explicit warmups	64 requests in four timing processes; 16 explicit warmups
Qualification	Eight reference requests and 12 independent qualification requests; 1,106+1,794 vectors	12 qualification requests; 346 saved PQ/QB full-logit pairs
Primary window	Fresh-context creation/reservation through prefill, generation, state/synchronization and teardown	Same declared service components; separate private driver and prefill policy
Excluded / separately charged	Shared model loading, post-timer comparison and raw-file I/O	Shared model loading, post-timer comparison and raw-file I/O

Table S21: All input-aggregate service ratios. Four sessions contribute to each row. A/A is the ratio of two identical PQ controls. Lower QB ratios favor QB. Inputs are different between platforms.

Platform	Input	Output / stop	QB / dual-PQ	QB / R	PQ_A / PQ_B
RTX	MBPP/413	34/EOG	0.988025	0.608126	0.947919
RTX	GSM8K/120	128/CAP	1.004279	0.578678	0.956661
RTX	GSM8K/322	128/CAP	0.963695	0.610027	0.990391
RTX	GSM8K/443	128/CAP	1.001663	0.535651	0.964986
RTX	**Equal-weight mean**	—	0.989284	0.582325	—
B570	HumanEval/38	64/CAP	0.994638	2.597826	0.999532
B570	HumanEval/65	53/EOG	0.992003	2.652017	0.997703
B570	HumanEval/115	41/EOG	0.997281	2.619785	0.999881
B570	HumanEval/127	64/CAP	0.996663	2.609733	0.992463
B570	**Equal-weight mean**	—	0.995144	2.619763	—

Table S22: Every input/session block from the two service panels. Each platform has its own input set; adjacent rows do not form cross-platform pairs. Exact source blocks are included in the ancillary CSV files.

RTX

s	Input	QB/PQ	QB/R	A/A
0	M/413	1.139826	0.860710	0.995684
0	G/120	0.945850	0.626761	0.948687
0	G/322	0.923908	0.620302	1.002893
0	G/443	0.922590	0.536318	0.937857
1	M/413	1.037841	0.605209	1.018199
1	G/120	1.060687	0.598746	0.928511
1	G/322	0.964679	0.643793	1.133151
1	G/443	1.095851	0.587714	0.908679
2	M/413	0.867052	0.507936	0.836860
2	G/120	1.075364	0.574851	0.996817
2	G/322	1.063244	0.656443	0.950265
2	G/443	1.031135	0.530409	1.012026
3	M/413	0.929090	0.516898	0.951652
3	G/120	0.942870	0.519815	0.953907
3	G/322	0.910152	0.528264	0.890920
3	G/443	0.965632	0.492411	1.005416

B570

s	Input	QB/PQ	QB/R	A/A
0	HE/38	0.994435	2.594996	1.000263
0	HE/65	0.994347	2.670002	0.998378
0	HE/115	0.997636	2.622364	0.999695
0	HE/127	0.998560	2.603504	1.001035
1	HE/38	0.995388	2.601387	0.997047
1	HE/65	0.990654	2.645055	1.001826
1	HE/115	0.997896	2.619781	1.000689
1	HE/127	0.997328	2.604889	0.998169
2	HE/38	0.997377	2.602636	1.001209
2	HE/65	0.992339	2.647094	0.997665
2	HE/115	0.996576	2.619385	0.998640
2	HE/127	0.998198	2.602524	0.998604
3	HE/38	0.991360	2.592299	0.999614
3	HE/65	0.990676	2.645998	0.992961
3	HE/115	0.997017	2.617613	1.000503
3	HE/127	0.992578	2.628099	0.972324

to a 41.8% service-time reduction; throughput uses the inverse time ratio. These four-input panels report descriptive paired estimates and every process-session block.

Warmups are separately recorded, but their presence alone establishes neither cold-driver behavior nor warmed state for all shapes. B570 uses the old 16-token key256 request for warmup; key512 pipeline persistence across processes is uncontrolled. The RTX panel warms each input/arm. Continuous exclusive device ownership is not established in either study. All blocks and reversals are re-

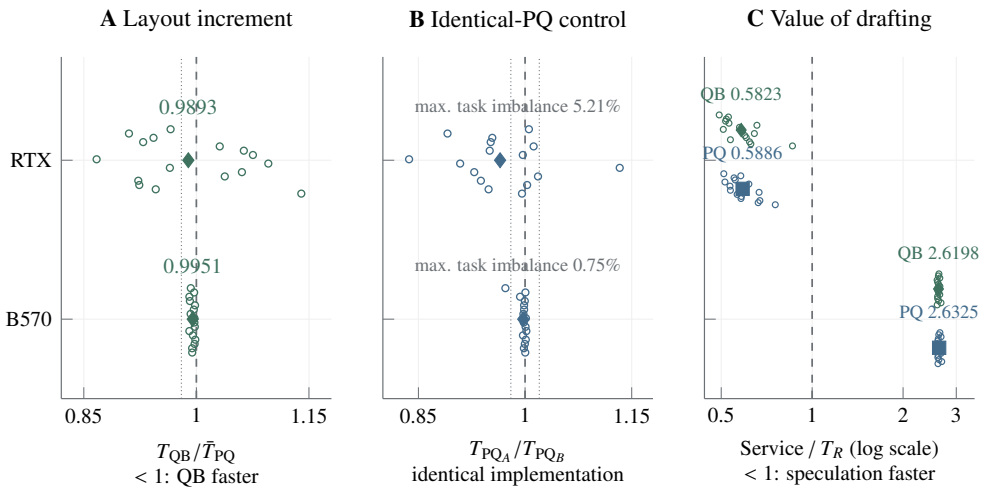


Figure S4: Service costs under separate platform contracts. Small outlined points show all 16 input/session blocks per contrast; large markers are equal-weight geometric means. A: QB/dual-PQ. B: identical-PQ A/A, with the largest absolute input-aggregate imbalance used by the screen. C: both speculative paths relative to target-only R. Dashed lines mark parity; dotted lines show 2%. Vertical offsets separate observations. Ratios below one favor the numerator. The platform panels are analyzed separately.

tained; no outlier deletion or significance-driven extension is used.

S8.3. Draft-policy counterfactual and completed prefill follow-up

The B570 thread-parameter study fixes the PR134 QB references and changes draft context parameters from 8 to 1 while keeping the target at 8. All 346 vectors and endpoints/state traces remain unchanged. Four timing processes yield 64 measurements plus 16 warmups. DRAFT1/dual-baseline service is 1.003994, DRAFT1/R is 2.633056 and the A/A criterion is 0.793%. The session ratios 0.998803, 1.007142, 1.010567 and 0.999512 do not support a practical improvement [17]. A temporary ggml pool object in an OpenMP build does not establish per-call creation of OS threads; the one-task GET_ROWS/SET_ROWS plan can clamp effective workers before the parameter takes effect.

The same reviewed commit also contains a draft-prefill submission follow-up. Its first attempt fails an observer-axis check; this failure is retained. A prospective observer-only repair checks the consumed QK tensor and all four inputs then qualify, with 346 matching logits, one draft-prefill API call and physical prefill $q = 1$. A later registered closeout completes this cost comparison under its separate CPU-draft service contract [18]. It records 156 requests: four calibration, 24 qualification, 64 matched warmups and 64 measurements. SUBMIT/dual-baseline is 0.999161, a 0.084% point reduction against 0.369% input-aggregate A/A and a 2% practical floor. One session favors the baseline. Fewer submissions therefore do not establish lower complete service cost, and QB/thread8 remains the baseline. SUBMIT/R is 1.203678 in this later contract; it cannot be compared with PR134's 2.619763 as a controlled improvement because device placement and service policy differ. The recorded helper-process reservation was not a complete observed count; full resource-protocol compliance is not established. The valid negative cost result and that accounting limitation are both retained.

S8.4. Evidence and execution accounting

The RTX study completes 12 native processes and 148 requests: eight references, 12 qualification, 64 measured and 64 warmups. All native exits and cleanup are recorded, with no scientific retry. The conditional 24 profiling slots remain unstarted under the registered stopping rule. Native parent lifetime is 446.746733 seconds; GPU-active time is unmeasured. The preparation staging failure is retained, and its earlier compiler lifetime was not captured.

The locally retained RTX archive contains 3,110 members and 2,900 vectors and is 2,270,281,687 bytes. The originating host reports reopening and hashing every member. The present manuscript package contains reported blocks, derived values, source identities and the original consistency inputs; it does not contain that full native archive. Local restoration, GitHub compact-record reconstruction and independent full-raw access are different evidence levels. B570's original full vectors also remain separately held.

S9. Matched deployment, input transfer and native reproduction

S9.1. Frozen matched design and uncertainty

The RTX bridge freezes at `30ad53be` before six formal process sessions. Its twelve-input panel is MBPP/123,140,255,401,413,418 and GSM8K/111,120,1308,141,322,443. Each session measures every input in four cells and three arms, giving $6 \times 12 \times 4 \times 3 = 864$ measurements; twelve warmup requests per session add 72. All 936 requests complete without formal retries, missing slots or outlier removal. Each cell occupies each order position three times per session. Each cell/session uses every three-arm permutation twice; each input/cell uses all six permutations across sessions.

The primary estimator is the exponential of the equal-input, equal-session mean of $\log[T_{QB}/\,\mathrm{mean}(T_{PQ_A}, T_{PQ_B})]$. The registered 20,000-resample bootstrap jointly resamples

the six session vectors, preserving the four paired cells. Each cell uses a 98.75% percentile interval, yielding nominal 95% family coverage across four primary cells. Practical success additionally requires an upper ratio below 0.98, a point reduction above both 2% and the input-aggregate A/A imbalance, and improvement in every session. With six sessions on one date, these intervals describe process-session variation on that host; date-level uncertainty is unmeasured.

The absolute arithmetic PQ/QB service means are 1.646575/1.631283 seconds for $k = 1$ resident, 1.358022/1.272156 for $k = 3$ resident, 1.690768/1.655382 for $k = 1$ fresh and 1.371483/1.307389 for $k = 3$ fresh. Ratios of these arithmetic means need not equal the primary geometric paired estimate. All per-input regressions, session signs, omit-one-session, chronological and pointwise t sensitivities remain in the archived analysis. The secondary $k3/k1$ ratios of layout ratios are 0.950119 resident and 0.972137 fresh; lifecycle contrasts are 0.986740 at $k = 1$ and 1.009607 at $k = 3$, with pointwise intervals spanning one for both lifecycle contrasts. The interaction is exploratory.

S9.2. Preserved engineering attempts and observation cost

A 25-request pilot precedes qualification. The first qualification attempt completes 135 requests before a Windows aggregate-file replacement failure leaves 69 slots unexecuted. A recording-only correction creates a new executable identity, and all 204 qualification requests are rerun before freezing formal timing. No formal samples are pooled across builds. A counter pilot has an unresolved approximately 2.9% difference, so the additional counters and heavy callback/logit capture are off during formal timing. Common driver bookkeeping, phase timers and inherited backend logging remain included. Actual CUDA capture/replay/fallback activity is not fully recorded.

S9.3. Previously unexposed panel and stopped families

The input-transfer study is separately frozen and does not share the bridge's exposed-input claim. It reaches 33 of 36 family/input cells, representing 20 of 21 distinct tasks. Independent compact reconstruction reconciles 1,018 completed and 758 unexecuted request slots; the two qualified families contribute 384 measurements and 384 warmups. No successful Llama subset is timed after its reference counterexample. Initial missing raw-offset metadata is retained; a heavy-observer correction and causal rules are fixed before held-out selection. The native-fast path fails old calibration for SmolLM2/Llama and held-out GSM8K/70 for Qwen3. Source-pinned counterexample token IDs, dispositions and all cost rows accompany the paper.

S9.4. Native evidence and access

RTX restores 3,542 original and 3,537 metadata4 source members, rebuilds core/frontend objects, and retains the pinned numerical DLLs. A real Qwen3 layer0/q3 fixture passes original/rebuilt MQ/PQ/QB replay with two repeats; six original/rebuilt HTTP R/PQ/QB requests match the historical endpoint. The subsequent delivery package exposes the real fixture, a complete natural-request R/PQ/QB panel, formal records and dependency acquisition instructions; the full 38,078-file raw directory remains separately held.

B570 rebuilds all five numerical libraries and 147 shader translation units from source. A new HumanEval/115 layer0/q2/key512 capture and its first-query q1 slice pass 16 stage comparisons. The ordinary four-input R/PQ/QB panel matches 914 historical complete vectors plus outputs, stops and state; copied-directory execution passes again. A diagnostic capture/supervision correction retains identical numerical DLL hashes. All 21 model requests and two stage workers are accounted for; nine additional cumulative Job processes lack identities/lifetimes. These resource limits do not change the recorded numerical comparisons. The full 1.37 GB compressed archive remains on the source host. The subsequent delivery publishes the source/stage/request subset and all four inputs' complete reference vectors as authenticated release assets. This reproduces PR134's two-model Vulkan contract, not PR143's later CPU-draft service policy.

`anc/followup_20261026/` carries exact compact records, source pins, derived tables and a read-only reconstruction path. The present manuscript revision performs CPU analysis only. Native build and replay results above are the platform executions recorded in the cited campaigns.

S10. Reproduction and evidence access

The source package builds the article and supplement from `main.tex`. It includes editable figures, reported cost/state matrices, original RTX session tables, later service blocks and recovered E2 request times. The later transfer, prefill and matched-deployment records are pinned separately (Supplement S9). Analysis scripts reconstruct the registered estimators, absolute request costs, repair–draft contrasts and session sensitivity. Source hashes bind copied inputs to their campaign records. Original derived summaries permit one adjacent binary64 value for platform rounding; the later reconstruction uses an absolute tolerance of 10^{-12}. Source bytes, integer counts and generated TeX remain exact.

`anc/runtime_reproduction/` supplies CPU interface tests and archived implementation sources/patches with provenance. These expose the query/mask mapping, phase/cache contract and scoped AV repair. The companion native kits now add build routes, real fixtures, replay commands and execution receipts for both platforms [6, 8]. RTX rebuilds the core and frontend from authenticated source while retaining pinned CUDA/BLAS and ggml numerical libraries. Its real $q = 3$ fixture reproduces original and rebuilt QK/softmax/AV bytes, and complete HTTP requests reproduce the historical endpoints. B570 rebuilds all five numerical libraries and 147 shader translation units without old object directories. Its real $q = 2$ stage fixture and four-input R/PQ/QB panel reproduce the declared stage bytes and 914 corresponding historical full-logit vectors. Both kits preserve new executable identities separately from historical binaries.

`anc/source_manifest.json` pins each campaign. Original RTX evidence is sealed at `f997262d`, the B570

support audit at `9fafc4ba`, RTX service at `9868acfb`, and B570 service/policy at `51febadd/e44c4df0`. The delivery closeout supplies the minimal source, fixture and full-request reference bytes through authenticated repository access [38, 39]. RTX distributes a 12-part archive, relocatable wrappers and verified acquisition of 21 pinned numerical DLLs. Its fresh-download CPU audit checks 785 indexed payload files and preserved R/PQ/QB records containing 704 vectors. B570 publishes separate minimal-source/request and four-input-reference archives. Their download audits authenticate all listed members, compare 175 preserved capture/reference vectors for one request panel, and check 914 stored reference vectors across all four inputs. The latter uses the references themselves to check internal consistency; it is distinct from the earlier native capture/reference comparison.

Both kits pin exact public model files and document runtime preparation. The access index `REPRODUCTION_ACCESS.md` provides immutable commit links, asset identities, hashes and commands; `anc/native_delivery/` retains compact delivery records. Downloads were checked on the source hosts under authorized accounts. They establish material delivery, while execution on a newly provisioned host requires native qualification. B570 records its historical Intel driver version; a fixed installer for that version is not supplied. Larger original raw archives remain in source-host custody. Reported-data reconstruction in the manuscript package works independently of those archives.

The later studies add recorded native execution on RTX and B570; manuscript reconstruction uses their saved measurements and does not rerun those experiments. `anc/historical_disposition.csv` and `anc/editorial_archive/` retain the 28 legacy groups and later failures. The archived `dae5aea9` AV model remains a same-policy parity control, with no hardware-performance claim. Generated TeX resides in `tables/`, outside `anc/`, so article compilation does not depend on ancillary-data handling. The README gives the check/build commands and separates CPU validation from native runtime requirements.